

Ch 12 Unsupervised Learning

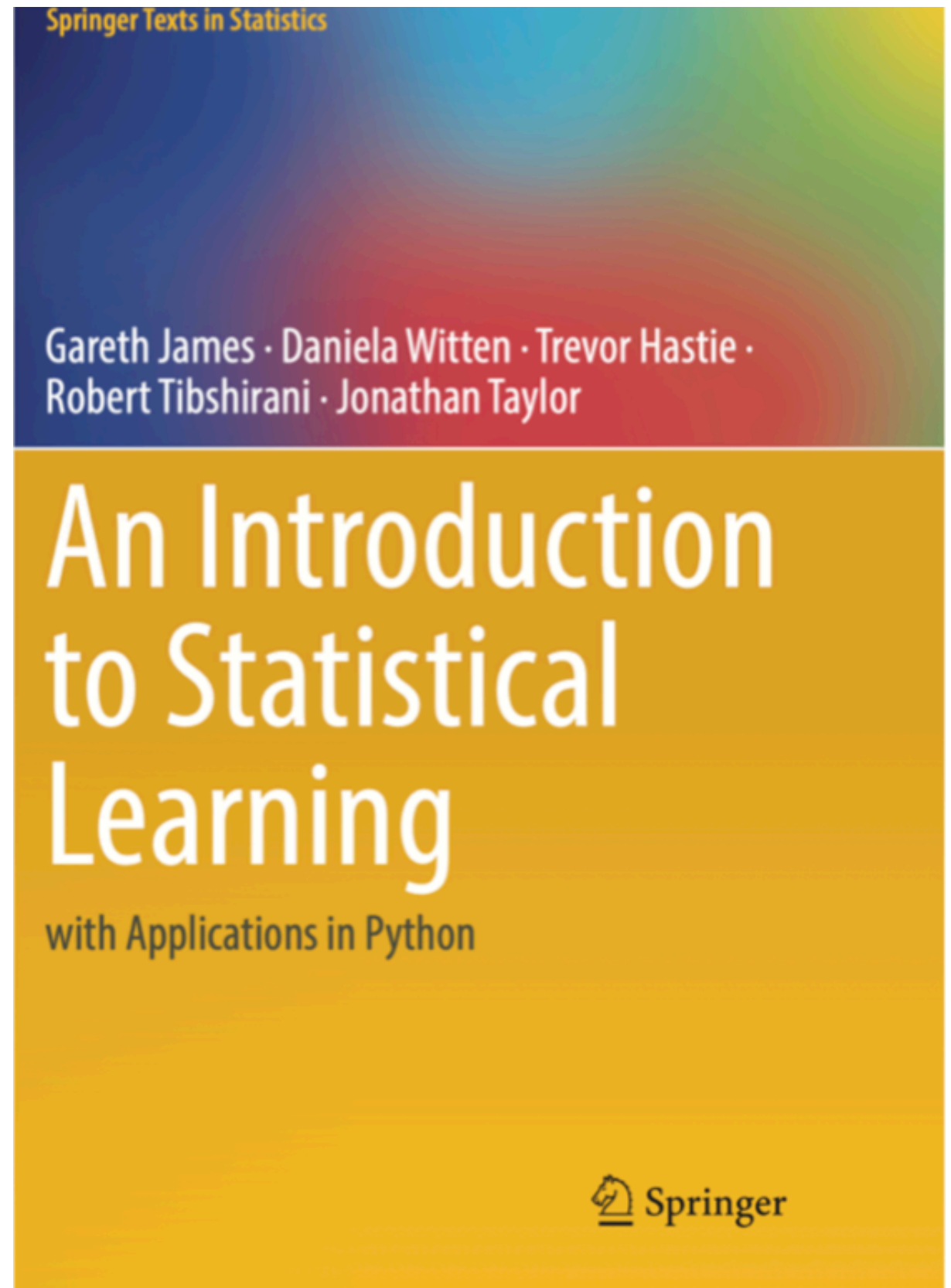
Instructor: Huei-Wen Teng

Slides modified from <https://www.statlearning.com/>

October 6, 2023

12.1

The Challenge of Unsupervised Learning



<https://www.statlearning.com/>

Unsupervised Learning

Unsupervised vs Supervised Learning:

- Most of this course focuses on *supervised learning* methods such as regression and classification.
- In that setting we observe both a set of features $X_1, X_2, \dots, X_p$ for each object, as well as a response or outcome variable Y . The goal is then to predict Y using $X_1, X_2, \dots, X_p$.
- Here we instead focus on *unsupervised learning*, where we observe only the features $X_1, X_2, \dots, X_p$. We are not interested in prediction, because we do not have an associated response variable Y .

The Goals of Unsupervised Learning

- The goal is to discover interesting things about the measurements: is there an informative way to visualize the data? Can we discover subgroups among the variables or among the observations?
- We discuss two methods:
 - *principal components analysis*, a tool used for data visualization or data pre-processing before supervised techniques are applied, and
 - *clustering*, a broad class of methods for discovering unknown subgroups in data.

The Challenge of Unsupervised Learning

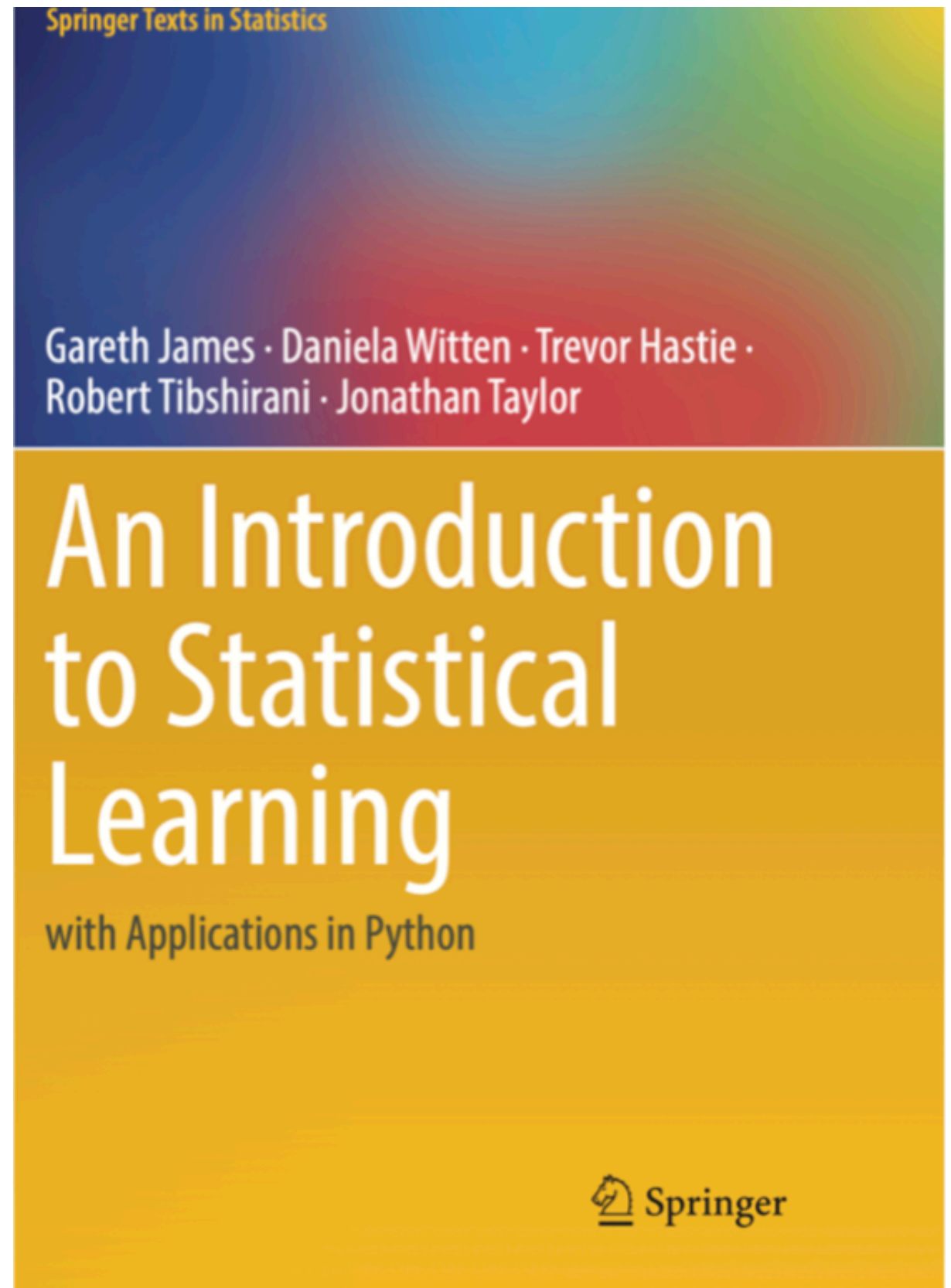
- Unsupervised learning is more subjective than supervised learning, as there is no simple goal for the analysis, such as prediction of a response.
- But techniques for unsupervised learning are of growing importance in a number of fields:
 - subgroups of breast cancer patients grouped by their gene expression measurements,
 - groups of shoppers characterized by their browsing and purchase histories,
 - movies grouped by the ratings assigned by movie viewers.

Another advantage

- It is often easier to obtain *unlabeled data* — from a lab instrument or a computer — than *labeled data*, which can require human intervention.
- For example it is difficult to automatically assess the overall sentiment of a movie review: is it favorable or not?

12.2

Principal Component Analysis



<https://www.statlearning.com/>

Principal Components Analysis

- PCA produces a low-dimensional representation of a dataset. It finds a sequence of linear combinations of the variables that have maximal variance, and are mutually uncorrelated.
- Apart from producing derived variables for use in supervised learning problems, PCA also serves as a tool for data visualization.

Principal Components Analysis: details

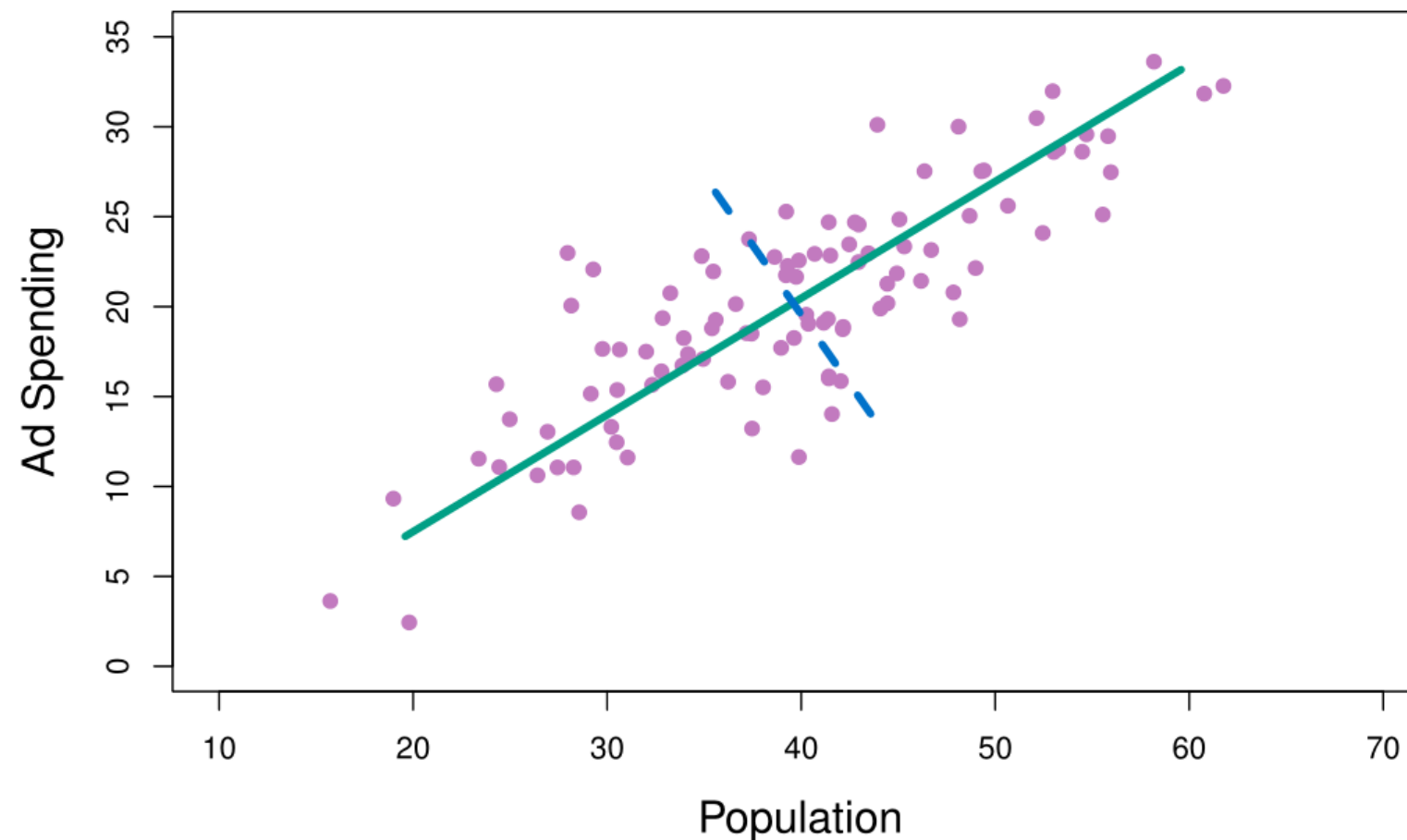
- The *first principal component* of a set of features $X_1, X_2, \dots, X_p$ is the normalized linear combination of the features

$$Z_1 = \phi_{11}X_1 + \phi_{21}X_2 + \dots + \phi_{p1}X_p$$

that has the largest variance. By *normalized*, we mean that $\sum_{j=1}^p \phi_{j1}^2 = 1$.

- We refer to the elements $\phi_{11}, \dots, \phi_{p1}$ as the loadings of the first principal component; together, the loadings make up the principal component loading vector, $\phi_1 = (\phi_{11} \ \phi_{21} \ \dots \ \phi_{p1})^T$.
- We constrain the loadings so that their sum of squares is equal to one, since otherwise setting these elements to be arbitrarily large in absolute value could result in an arbitrarily large variance.

PCA: example



The population size (**pop**) and ad spending (**ad**) for 100 different cities are shown as purple circles. The green solid line indicates the first principal component direction, and the blue dashed line indicates the second principal component direction.

Computation of Principal Components

- Suppose we have a $n \times p$ data set $\mathbf{X}$. Since we are only interested in variance, we assume that each of the variables in $\mathbf{X}$ has been centered to have mean zero (that is, the column means of $\mathbf{X}$ are zero).
- We then look for the linear combination of the sample feature values of the form

$$z_{i1} = \phi_{11}x_{i1} + \phi_{21}x_{i2} + \dots + \phi_{p1}x_{ip} \quad (1)$$

for $i = 1, \dots, n$ that has largest sample variance, subject to the constraint that $\sum_{j=1}^p \phi_{j1}^2 = 1$.

- Since each of the x_{ij} has mean zero, then so does z_{i1} (for any values of ϕ_{j1}). Hence the sample variance of the z_{i1} can be written as $\frac{1}{n} \sum_{i=1}^n z_{i1}^2$.

Computation: continued

- Plugging in (1) the first principal component loading vector solves the optimization problem

$$\underset{\phi_{11}, \dots, \phi_{p1}}{\text{maximize}} \frac{1}{n} \sum_{i=1}^n \left(\sum_{j=1}^p \phi_{j1} x_{ij} \right)^2 \quad \text{subject to} \quad \sum_{j=1}^p \phi_{j1}^2 = 1.$$

- This problem can be solved via a singular-value decomposition of the matrix $\mathbf{X}$, a standard technique in linear algebra.
- We refer to Z_1 as the first principal component, with realized values $z_{11}, \dots, z_{n1}$

Geometry of PCA

- The loading vector ϕ_1 with elements $\phi_{11}, \phi_{21}, \dots, \phi_{p1}$ defines a direction in feature space along which the data vary the most.
- If we project the n data points $x_1, \dots, x_n$ onto this direction, the projected values are the principal component scores $z_{11}, \dots, z_{n1}$ themselves.

Further principal components

- The second principal component is the linear combination of $X_1, \dots, X_p$ that has maximal variance among all linear combinations that are *uncorrelated* with Z_1 .
- The second principal component scores $z_{12}, z_{22}, \dots, z_{n2}$ take the form

$$z_{i2} = \phi_{12}x_{i1} + \phi_{22}x_{i2} + \dots + \phi_{p2}x_{ip},$$

where ϕ_2 is the second principal component loading vector, with elements $\phi_{12}, \phi_{22}, \dots, \phi_{p2}$.

Further principal components: continued

- It turns out that constraining Z_2 to be uncorrelated with Z_1 is equivalent to constraining the direction ϕ_2 to be orthogonal (perpendicular) to the direction ϕ_1 . And so on.
- The principal component directions $\phi_1, \phi_2, \phi_3, \dots$ are the ordered sequence of right singular vectors of the matrix $\mathbf{X}$, and the variances of the components are $\frac{1}{n}$ times the squares of the singular values. There are at most $\min(n - 1, p)$ principal components.

Illustration

- **USAarrests** data: For each of the fifty states in the United States, the data set contains the number of arrests per 100,000 residents for each of three crimes: **Assault**, **Murder**, and **Rape**. We also record **UrbanPop** (the percent of the population in each state living in urban areas).
- The principal component score vectors have length $n = 50$, and the principal component loading vectors have length $p = 4$.
- PCA was performed after standardizing each variable to have mean zero and standard deviation one.

USAarrests data: PCA plot

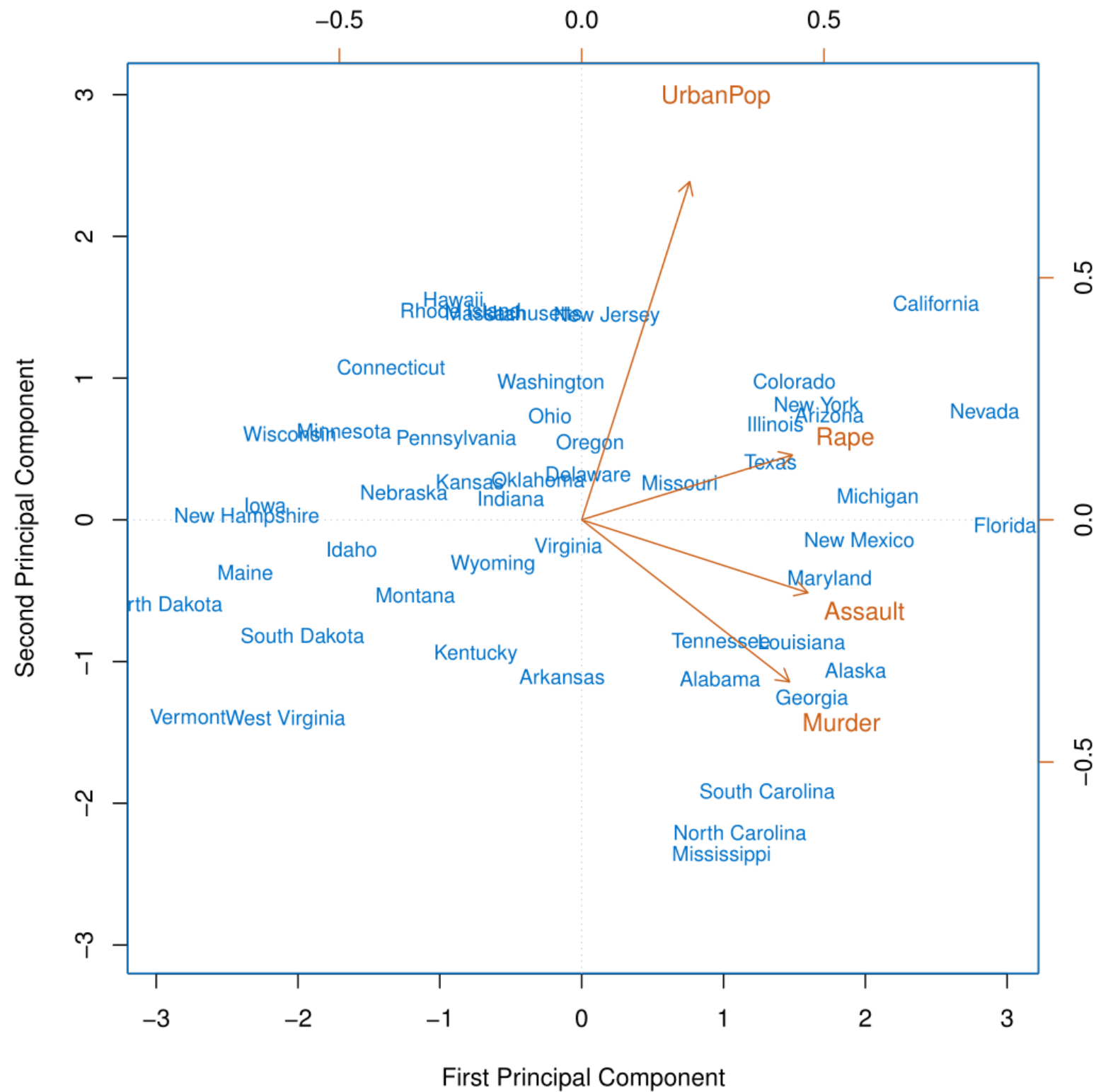


Figure details

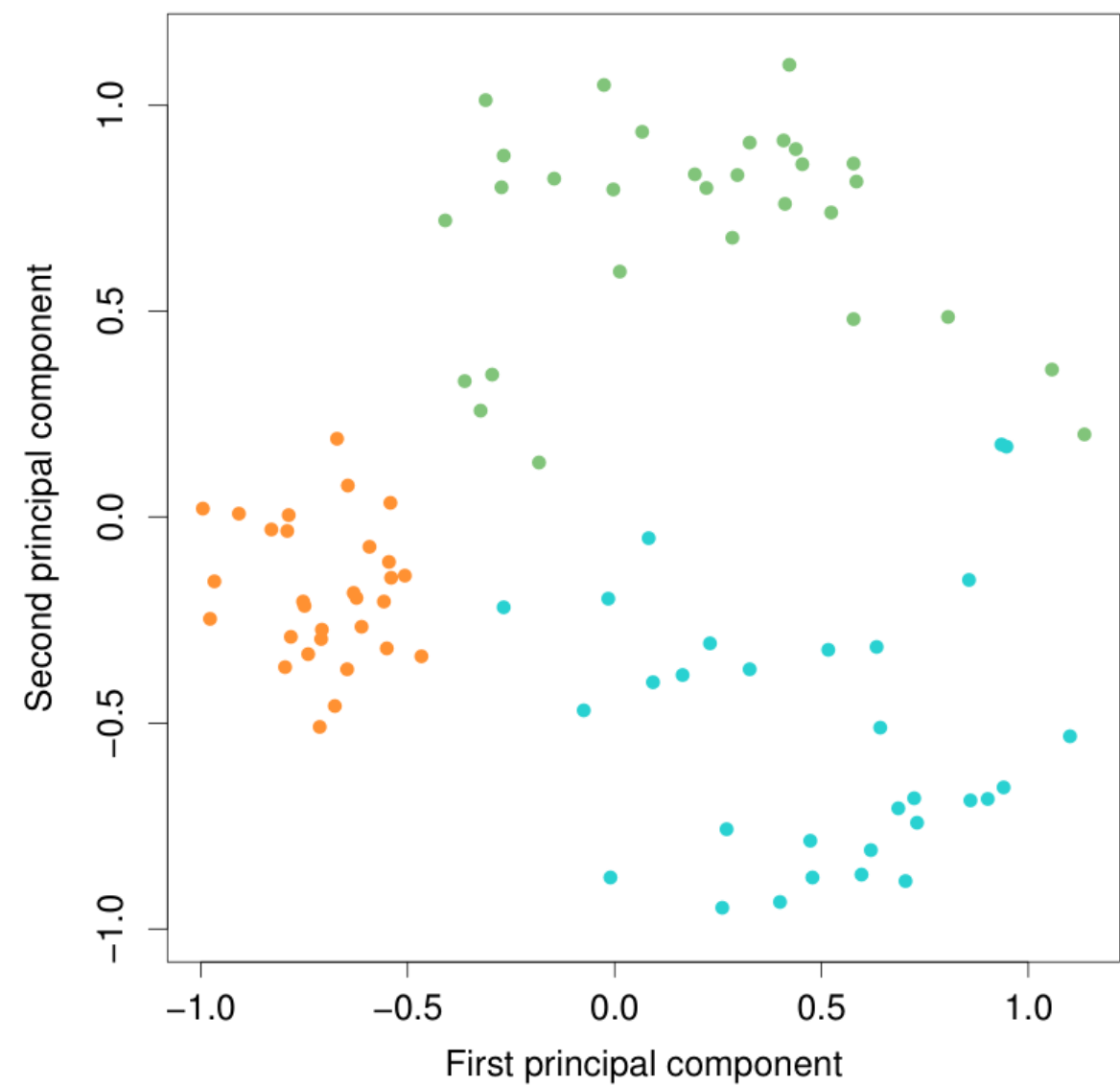
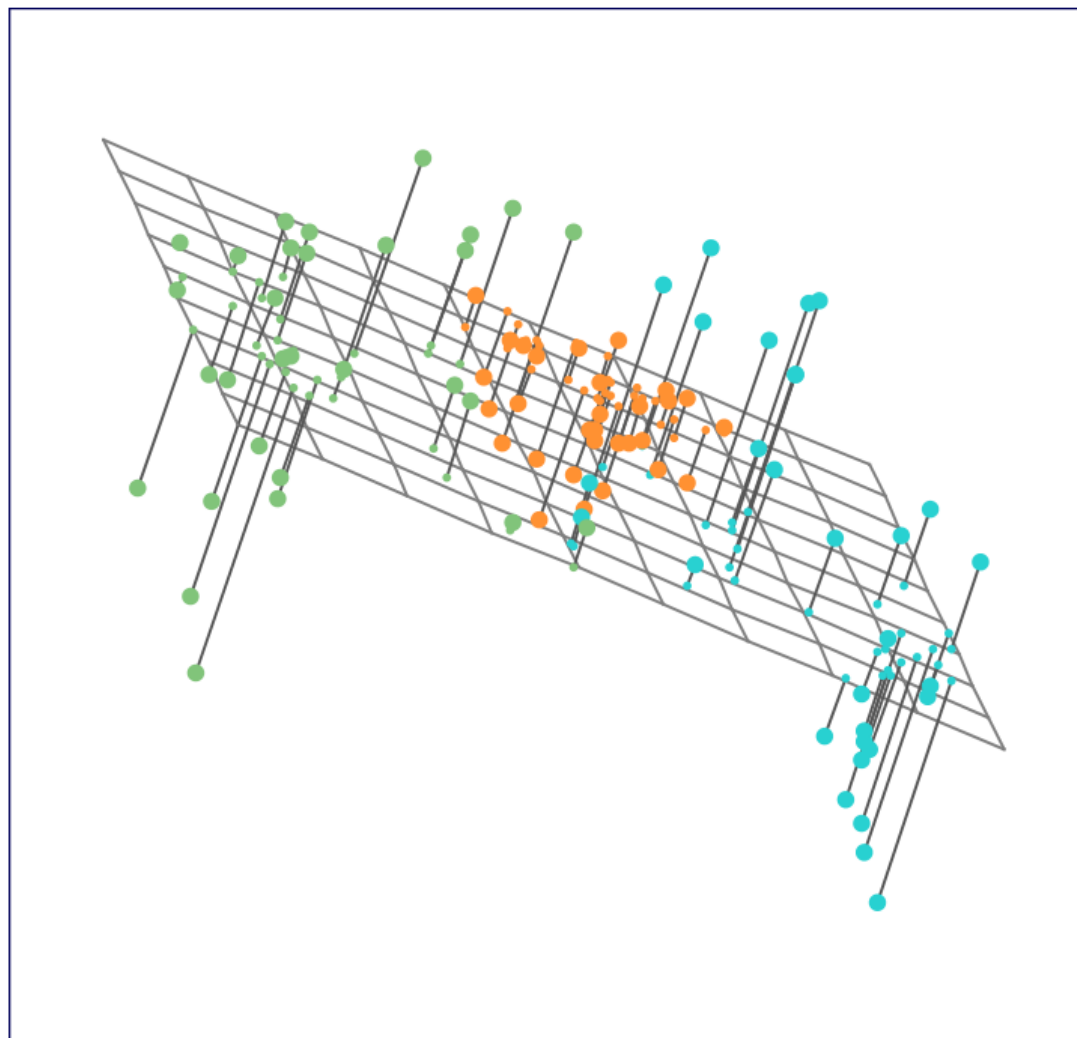
The first two principal components for the USArrests data.

- The blue state names represent the scores for the first two principal components.
- The orange arrows indicate the first two principal component loading vectors (with axes on the top and right). For example, the loading for **Rape** on the first component is 0.54, and its loading on the second principal component 0.17 [the word **Rape** is centered at the point (0.54, 0.17)].
- This figure is known as a *biplot*, because it displays both the principal component scores and the principal component loadings.

PCA loadings

	PC1	PC2
Murder	0.5358995	-0.4181809
Assault	0.5831836	-0.1879856
UrbanPop	0.2781909	0.8728062
Rape	0.5434321	0.1673186

Another Interpretation of Principal Components

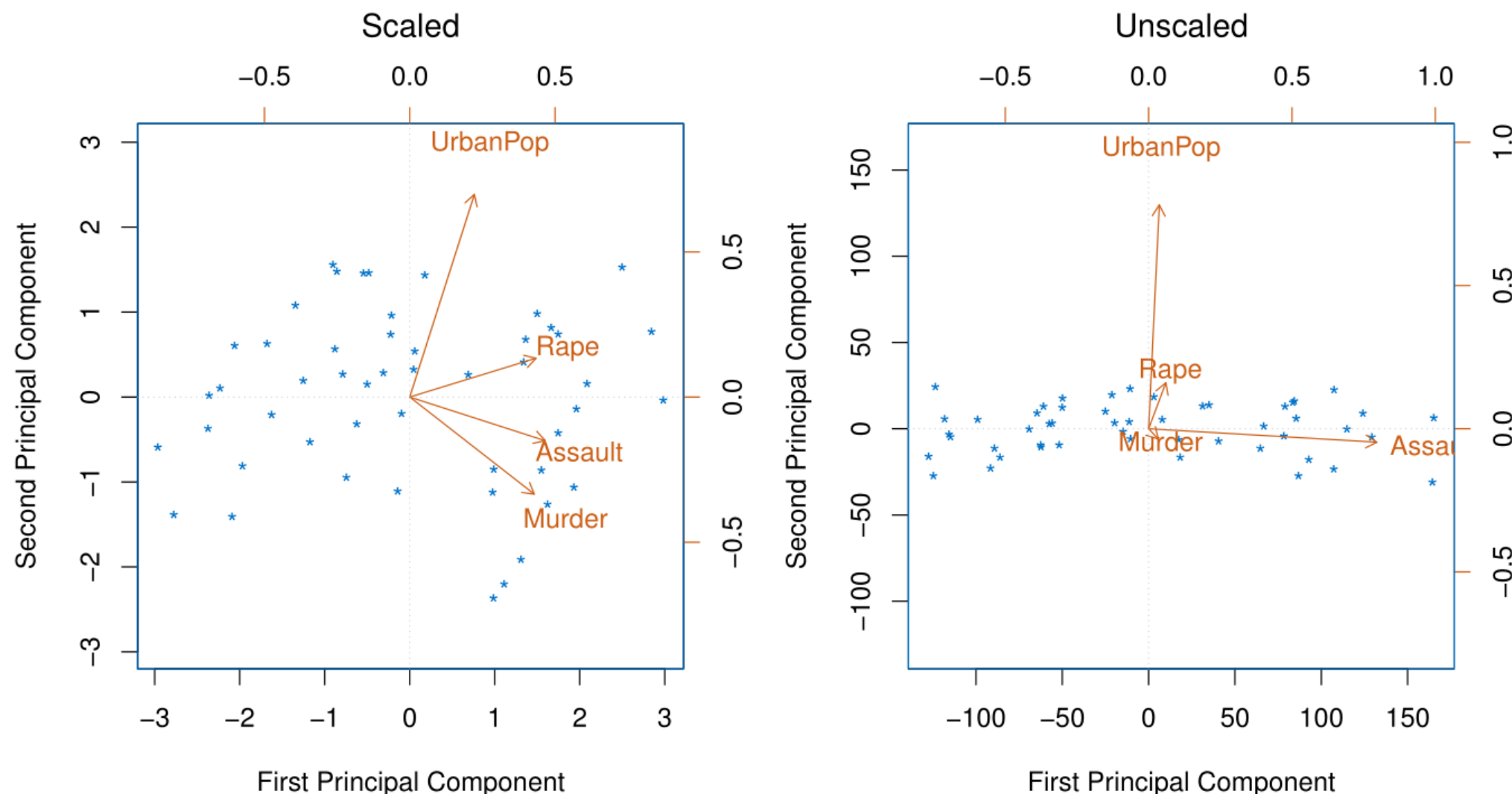


PCA find the hyperplane closest to the observations

- The first principal component loading vector has a very special property: it defines the line in p -dimensional space that is *closest* to the n observations (using average squared Euclidean distance as a measure of closeness)
- The notion of principal components as the dimensions that are closest to the n observations extends beyond just the first principal component.
- For instance, the first two principal components of a data set span the plane that is closest to the n observations, in terms of average squared Euclidean distance.

Scaling of the variables matters

- If the variables are in different units, scaling each to have standard deviation equal to one is recommended.
- If they are in the same units, you might or might not scale the variables.



Proportion Variance Explained

- To understand the strength of each component, we are interested in knowing the proportion of variance explained (PVE) by each one.
- The *total variance* present in a data set (assuming that the variables have been centered to have mean zero) is defined as

$$\sum_{j=1}^p \text{Var}(X_j) = \sum_{j=1}^p \frac{1}{n} \sum_{i=1}^n x_{ij}^2,$$

and the variance explained by the m th principal component is

$$\text{Var}(Z_m) = \frac{1}{n} \sum_{i=1}^n z_{im}^2.$$

- It can be shown that $\sum_{j=1}^p \text{Var}(X_j) = \sum_{m=1}^M \text{Var}(Z_m)$, with $M = \min(n - 1, p)$.

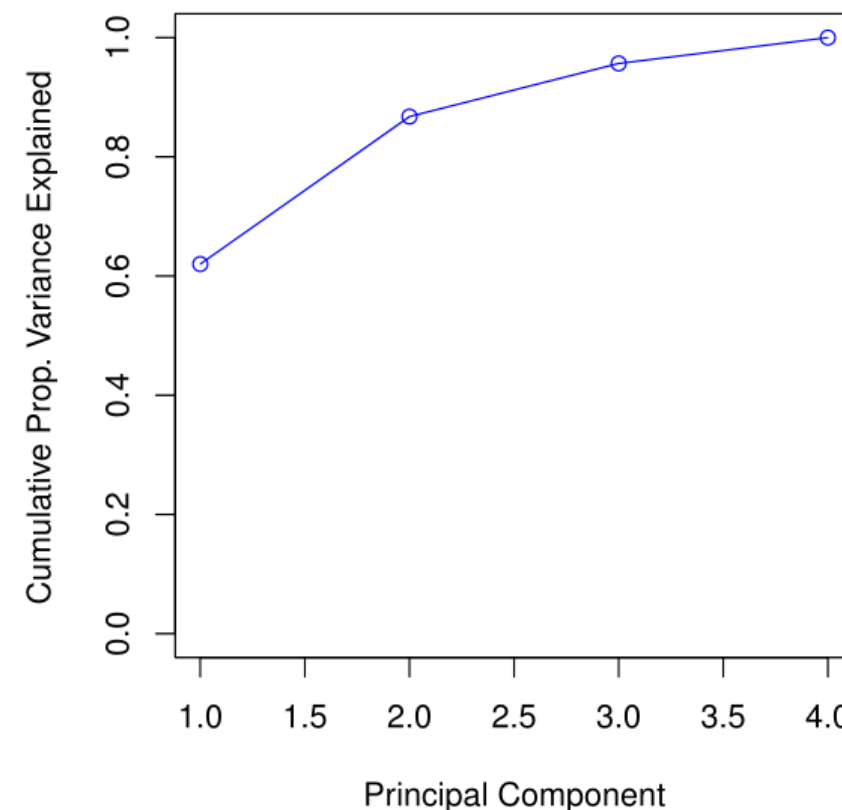
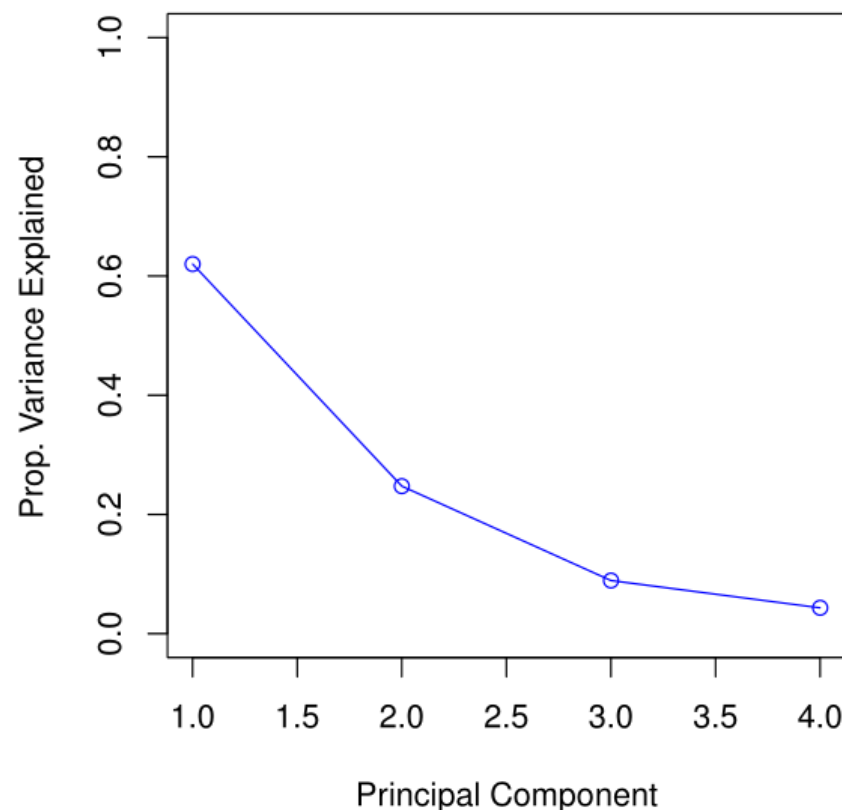
Proportion Variance Explained: continued

- Therefore, the PVE of the m th principal component is given by the positive quantity between 0 and 1

$$\frac{\sum_{i=1}^n z_{im}^2}{\sum_{j=1}^p \sum_{i=1}^n x_{ij}^2}.$$

- The PVEs sum to one. We sometimes display the cumulative PVEs.

Scree plot



How many principal components should we use?

If we use principal components as a summary of our data, how many components are sufficient?

- No simple answer to this question, as cross-validation is not available for this purpose.
 - *Why not?*
 - When could we use cross-validation to select the number of components?
- the “scree plot” on the previous slide can be used as a guide: we look for an “elbow”.

▣ See 20261005-PCA-wine.ipynb 

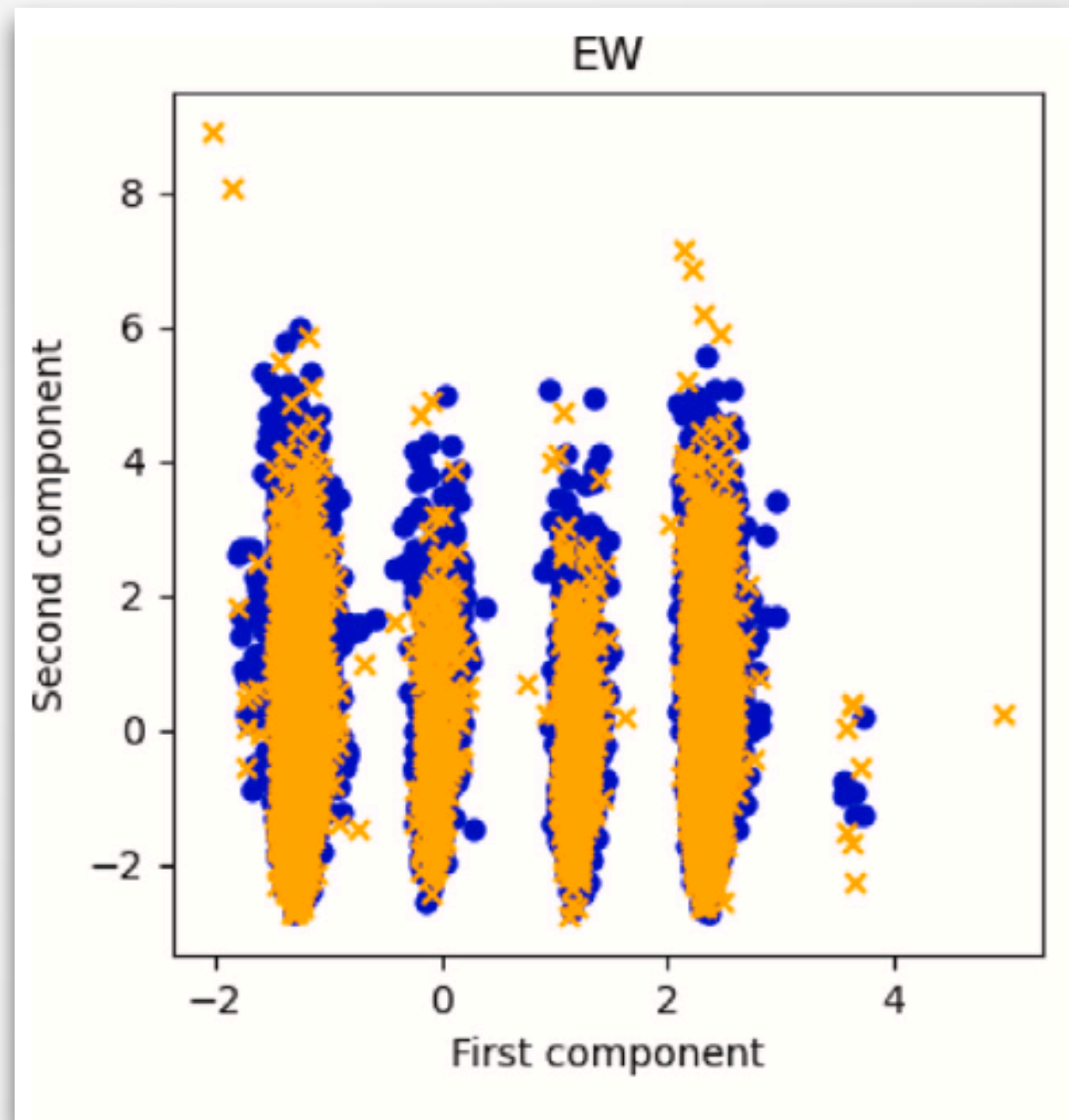
When to use PCA?

- **Visualization:** in two or three dimensions
- **PCA regression**
 - Usually, explanatory variables are highly correlated.
 - Feature engineering: Use PCA to create a new set of variables (which are mainly linear combinations of the original variables).

Visualization

- Dimension reduction for visualization
- Insights? Rescaled Cluster-then-predict

Teng et al (2024)



<https://www.sciencedirect.com/science/article/abs/pii/S1057521923005215>



PCA regression

- ▣ Feature engineering
 - ▶ Find representative features
 - ▶ Avoid multicollinearity problems
 - ▶ Improve prediction

Tuan et al (2023)

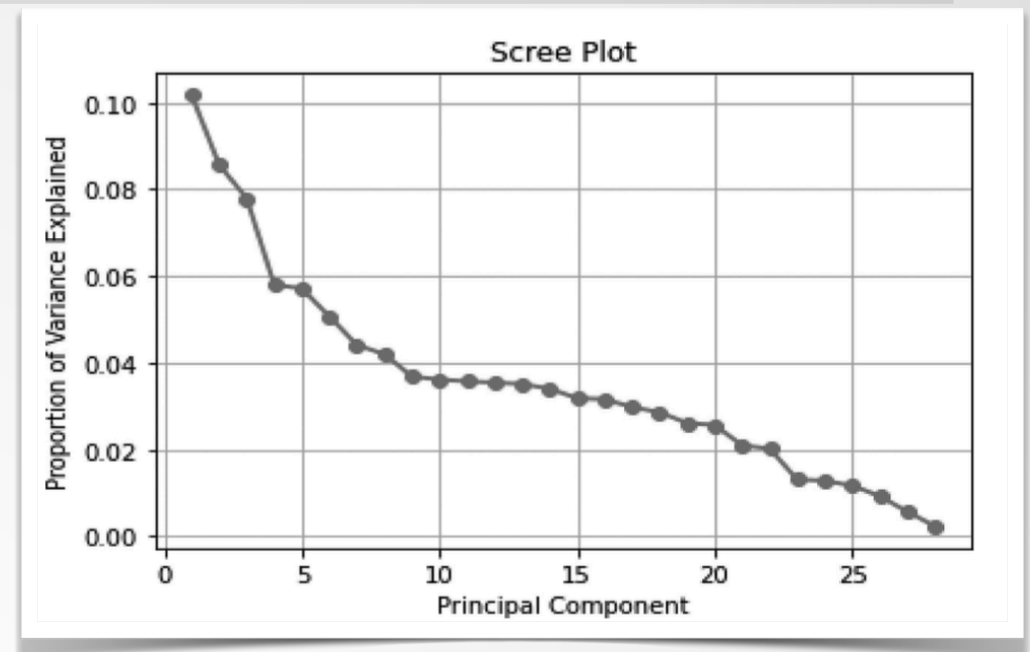


Table 3. Comparison of bank failure prediction methods: unadjusted data and after PCA.

Panel A: Unadjusted data				
In-sample test	Logistic regression	KNN	Decision Tree	Random Forest
Accuracy	0.6325	0.9459	1.0000	1.0000
Precision	0.6261	0.9422	1.0000	1.0000
Recall	0.4625	0.9378	1.0000	1.0000
Out-of-sample test				
Accuracy	0.6125	0.5689	0.5111	0.7076
Precision	0.2109	0.1847	0.1574	0.2389
Recall	0.5482	0.5227	0.4961	0.4070
Panel B: After PCA				
In-sample test				
Accuracy	0.6169	0.8578	1.0000	1.0000
Precision	0.6154	0.8544	1.0000	1.0000
Recall	0.4050	0.8260	1.0000	1.0000
Out-of-sample test				
Accuracy	0.5154	0.8109	0.7005	0.8057
Precision	0.1563	0.4372	0.2964	0.4278
Recall	0.4845	0.7735	0.6805	0.7558



PCA vs Clustering

- PCA looks for a low-dimensional representation of the observations that explains a good fraction of the variance.
- Clustering looks for homogeneous subgroups among the observations.



Machine Learning and FinTech: PCA more

Huei-Wen Teng

National Yang Ming Chiao Tung University

A quick look at linear algebra and basis transformation

1. Standard vs. new basis

- Standard basis:

$$e_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad e_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

- New orthonormal basis (rotated 45°):

$$u_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad u_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

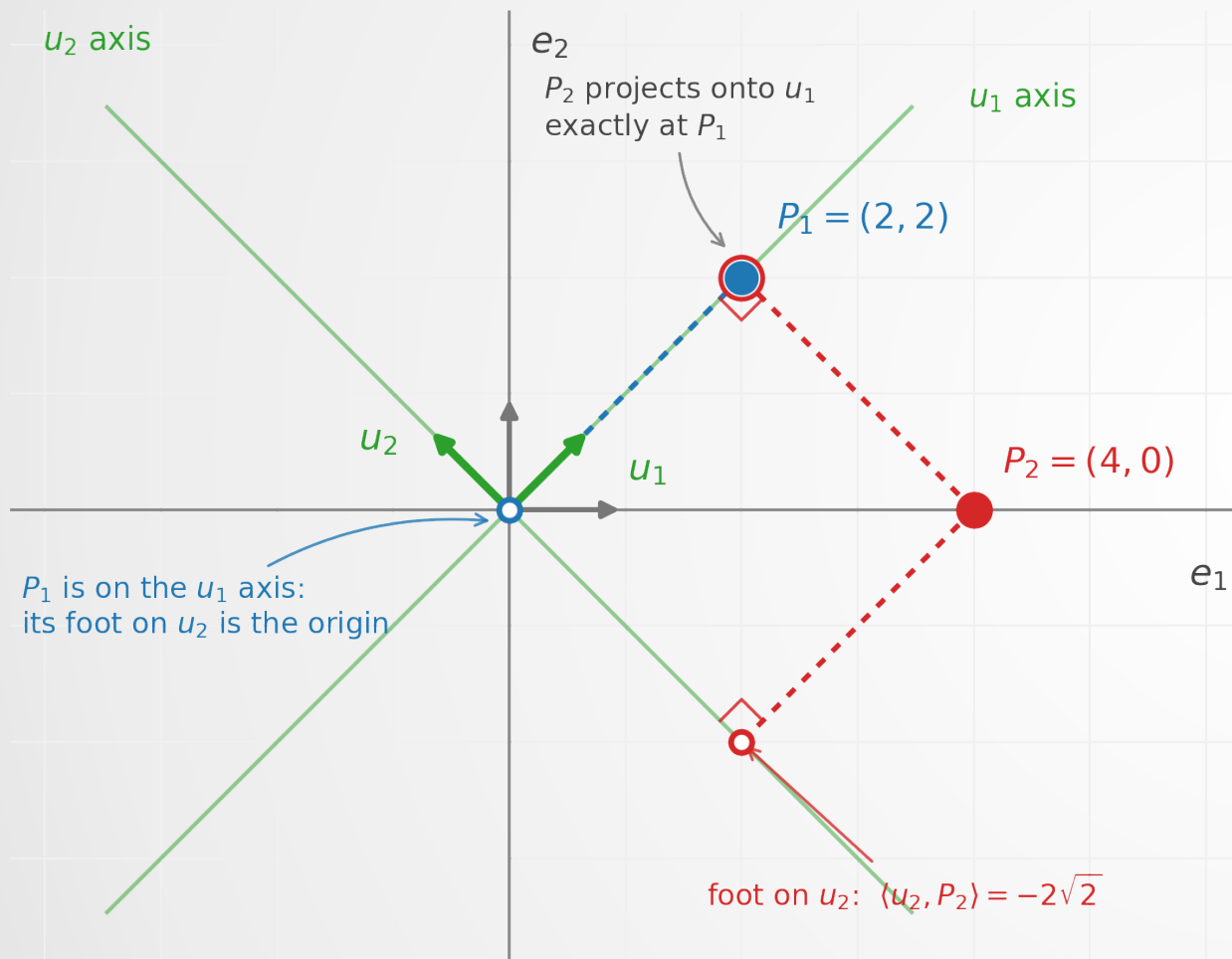
2. Points in the plane

$$P_1 = \begin{bmatrix} 2 \\ 2 \end{bmatrix}, \quad P_2 = \begin{bmatrix} 4 \\ 0 \end{bmatrix}.$$

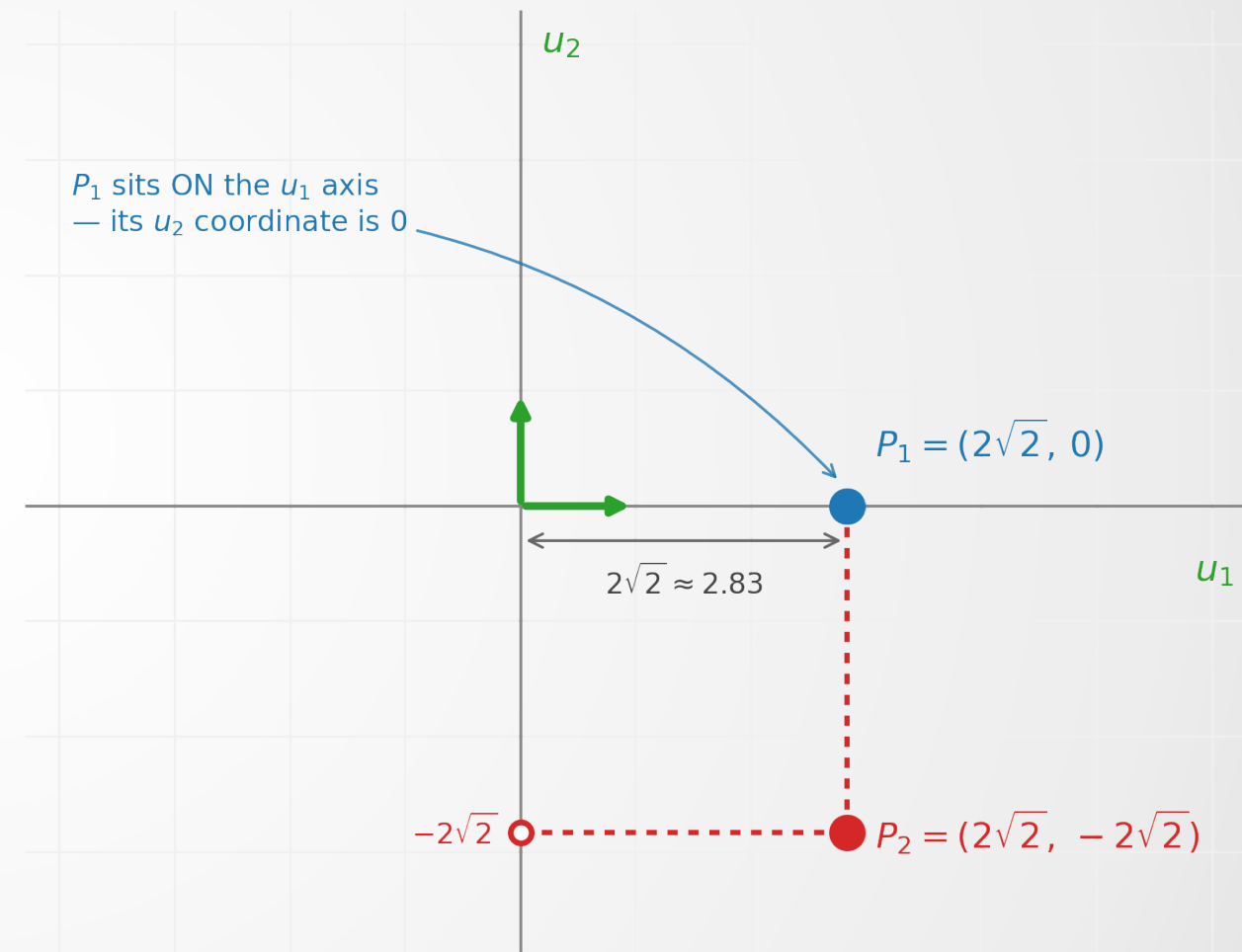


Change of basis: the points do not move — only their coordinates do

Original coordinates, basis $\{e_1, e_2\}$
the new directions u_1, u_2 drawn in



New coordinates, basis $\{u_1, u_2\}$
 $c_1 = \langle u_1, P \rangle, c_2 = \langle u_2, P \rangle$



3. Coordinates in the new basis

Since $\{u_1, u_2\}$ is an **orthonormal basis**, the coordinates are simply the **inner products**:

$$c_1 = \langle u_1, P \rangle, \quad c_2 = \langle u_2, P \rangle.$$

- For P_1 :

$$\langle u_1, P_1 \rangle = \frac{1}{\sqrt{2}}(2 + 2) = 2\sqrt{2}, \quad \langle u_2, P_1 \rangle = \frac{1}{\sqrt{2}}(-2 + 2) = 0.$$

So in the new basis:

$$P_1 = (2\sqrt{2}, 0).$$

- For P_2 :

$$\langle u_1, P_2 \rangle = \frac{1}{\sqrt{2}}(4 + 0) = 2\sqrt{2}, \quad \langle u_2, P_2 \rangle = \frac{1}{\sqrt{2}}(-4 + 0) = -2\sqrt{2}.$$

So in the new basis:

$$P_2 = (2\sqrt{2}, -2\sqrt{2}).$$

4. Interpretation

- The **inner product with u_1** gives the "shadow" of the point on the u_1 -axis.
- The **inner product with u_2** gives the "shadow" on the u_2 -axis.
- Together, they are the new coordinates in this rotated orthonormal system.
- The **points in space remain the same**, only their coordinates change with the chosen basis.



Notations: Column vector by default

$$\mathbf{X} = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \vdots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{pmatrix}$$

The i -th observation

$$x_i = \begin{pmatrix} x_{i1} \\ x_{i2} \\ \vdots \\ x_{in} \end{pmatrix}$$

The j -th feature

$$\mathbf{x}_j = \begin{pmatrix} x_{1j} \\ x_{2j} \\ \vdots \\ x_{nj} \end{pmatrix}$$

$$\mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_p)$$

$$\mathbf{X} = (x_1^\top, x_2^\top, \dots, x_n^\top)$$

$^\top$: vector or matrix transpose

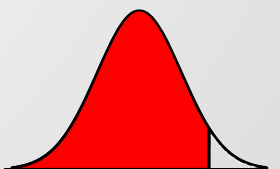


Notations

- ▣ The transpose of a vector $x_i^T = (x_{i1} \ x_{i2} \ \cdots \ x_{ip})$
- ▣ The transpose of a matrix

- ▣
$$\mathbf{X}^T = \begin{pmatrix} x_{11} & x_{21} & \cdots & x_{n1} \\ x_{12} & x_{22} & \cdots & x_{n2} \\ \vdots & \vdots & \vdots & \vdots \\ x_{1p} & x_{2p} & \cdots & x_{np} \end{pmatrix}$$

$$\mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$$



Preliminaries

- For a $p \times p$ matrix A , a non zero vector v , and a value λ .
- If $Av = \lambda v$, then v is an eigenvector of A with eigenvalue λ .
- If $\{v_1, \dots, v_p\}$ is a basis, such that v_i is eigenvector of A with eigenvalue λ_i . Then,

$$A \begin{pmatrix} v_1 & \cdots & v_p \end{pmatrix} = \begin{pmatrix} \lambda_1 v_1 & \cdots & \lambda_p v_p \end{pmatrix} = \begin{pmatrix} v_1 & \cdots & v_p \end{pmatrix} \begin{pmatrix} \lambda_1 & & \\ & \cdots & \\ & & \lambda_p \end{pmatrix}$$

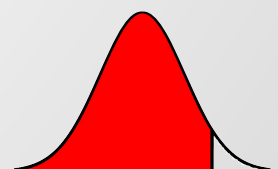
- Denote $V = \begin{pmatrix} v_1 & \cdots & v_p \end{pmatrix}$.

- We write $AV = V\Lambda$, or $A = V\Lambda V^{-1}$, and say A is diagonalizable.



Definitions

- ▣ A matrix is symmetric if $A^T = A$
- ▣ An orthonormal basis satisfies $V^{-1} = V^T$
 - ▣ $VV^T = VV^{-1} = I$
 - ▣ $v_i \perp v_j$ for $i \neq j$
 - ▣ $\|v_i\| = 1$ for all i



The Principal Axis Theorem

- When A is a real symmetric matrix, then A admits an orthonormal eigen basis.
- When A is a real and symmetric matrix, A can be diagonalized,
 $A = V\Lambda V^T$.

▶ $\Lambda = \begin{pmatrix} \lambda_1 & & \\ & \cdots & \\ & & \lambda_p \end{pmatrix}$ is the diagonal matrix with
 $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p$,

▶ $V = \begin{pmatrix} v_1 & \cdots & v_p \end{pmatrix}$ is the orthonormal matrix of Eigen vectors
 $v_1, \dots, v_p$, i.e., $V^T V = I$.



Data matrix $\mathbf{X}$

- For an $n \times p$ data matrix $\mathbf{X}$, $\mathbf{X}^T \mathbf{X}$ is real symmetric, because
$$(\mathbf{X}^T \mathbf{X})^T = \mathbf{X}^T (\mathbf{X}^T)^T = \mathbf{X}^T \mathbf{X}$$
- Thus, $\mathbf{X}^T \mathbf{X}$ is diagonalizable with orthonormal eigen basis.



PCA

- PCA actually seeks to find $w = (w_1, \dots, w_p)^T$ to maximize the variance of $Z_1 = \mathbf{X}w = X_1w_1 + X_2w_2 + \dots + X_pw_p$.
- PAC aims at $\max_{\|w\|^2=1} \text{var}(Z_1)$
- Because Z_1 has a column mean 0, its variance equals

$$\text{Var}(Z_1) = \frac{1}{n} \|Z_1\|^2 = \frac{1}{n} \|\mathbf{X}w\|^2$$

- PCA equals to $\max_{\|w\|^2=1} \|\mathbf{X}w\|^2$



□ However

$$\|\mathbf{X}w\|^2 = w^T \mathbf{X}^T \mathbf{X} w = w^T V \Lambda V^T w$$

$$= \tilde{w}^T \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_p \end{pmatrix} \tilde{w}$$

$$= \lambda_1 \tilde{w}_1^2 + \dots + \lambda_p \tilde{w}_p^2$$

$$\leq \lambda_1$$

$$V^T w = \begin{pmatrix} v_1^T \\ \vdots \\ v_p^T \end{pmatrix} \quad w = \tilde{w}$$

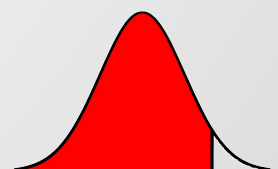


How do we do eigen decomposition for $\mathbf{X}^T \mathbf{X}$?

(1) Eigen decomposition

- The eigen decomposition procedure

$(\mathbf{X}^T \mathbf{X}) = V \Lambda V^{-1}$, where V is the matrix of eigenvectors, and Λ is the diagonal matrix with eigenvalues in the diagonal.



How do we do eigen decomposition for $\mathbf{X}^T \mathbf{X}$?

(2) Singular Value Decomposition (SVD)

The SVD of the $N \times p$ matrix X has the form $X = UDV'$.

- ▶ $U = (u_1, \dots, u_N)$ is an $N \times N$ orthogonal matrix. $\{u_1, \dots, u_N\}$ form an orthonormal basis for the space spanned by the column vectors of X .
- ▶ $V = (v_1, \dots, v_p)$ is an $p \times p$ orthogonal matrix. $\{v_1, \dots, v_p\}$ form an orthonormal basis for the space spanned by the row vector of X .
- ▶ D is an $N \times p$ rectangular matrix with nonzero elements along the first $p \times p$ submatrix diagonal. $\text{diag}(d_1, d_2, \dots, d_p)$, $d_1 \geq d_2 \geq \dots \geq d_p \geq 0$ are the singular values of X with $N > p$. Let plum denote matrix and vector transpose.

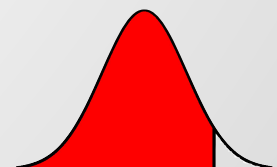


With SVD of X , we have

$$\begin{aligned} X'X &= (UDV')'(UDV') \\ &= VD'U'UDV' \\ &= VD'DV' \\ &= VD^2V'. \end{aligned}$$

Here, $D^2 = D'D$. If you have the SVP, you already have the Eigen value decomposition for $X'X$.

- ▶ The columns of V (i.e., $v_j, j = 1, \dots, p$) are the eigenvectors of $X'X$. They are called *principle component direction* of X .
- ▶ The diagonal values in D (i.e., $d_1, j = 1, \dots, p$) are the square roots of the eigenvalues of $X'X$.



A probabilistic view

- ▣ Suppose $\chi \sim N_p(\mathbf{0}, \Sigma)$. Find $w = (w_1, \dots, w_p)^\top$ satisfying $\|w\|^2 = 1$ to maximize $\text{Var}(w^\top \chi)$

$$\text{Var}(w^\top \chi) = w^\top \Sigma w$$

$$\approx \frac{1}{n} w^\top X^\top X w = \frac{1}{n} w^\top V \Lambda V^\top w$$

$$= \frac{1}{n} \tilde{w}^\top \Lambda \tilde{w}$$

$$= \frac{1}{n} (\lambda_1 \tilde{w}_1^2 + \dots + \lambda_p \tilde{w}_p^2)$$

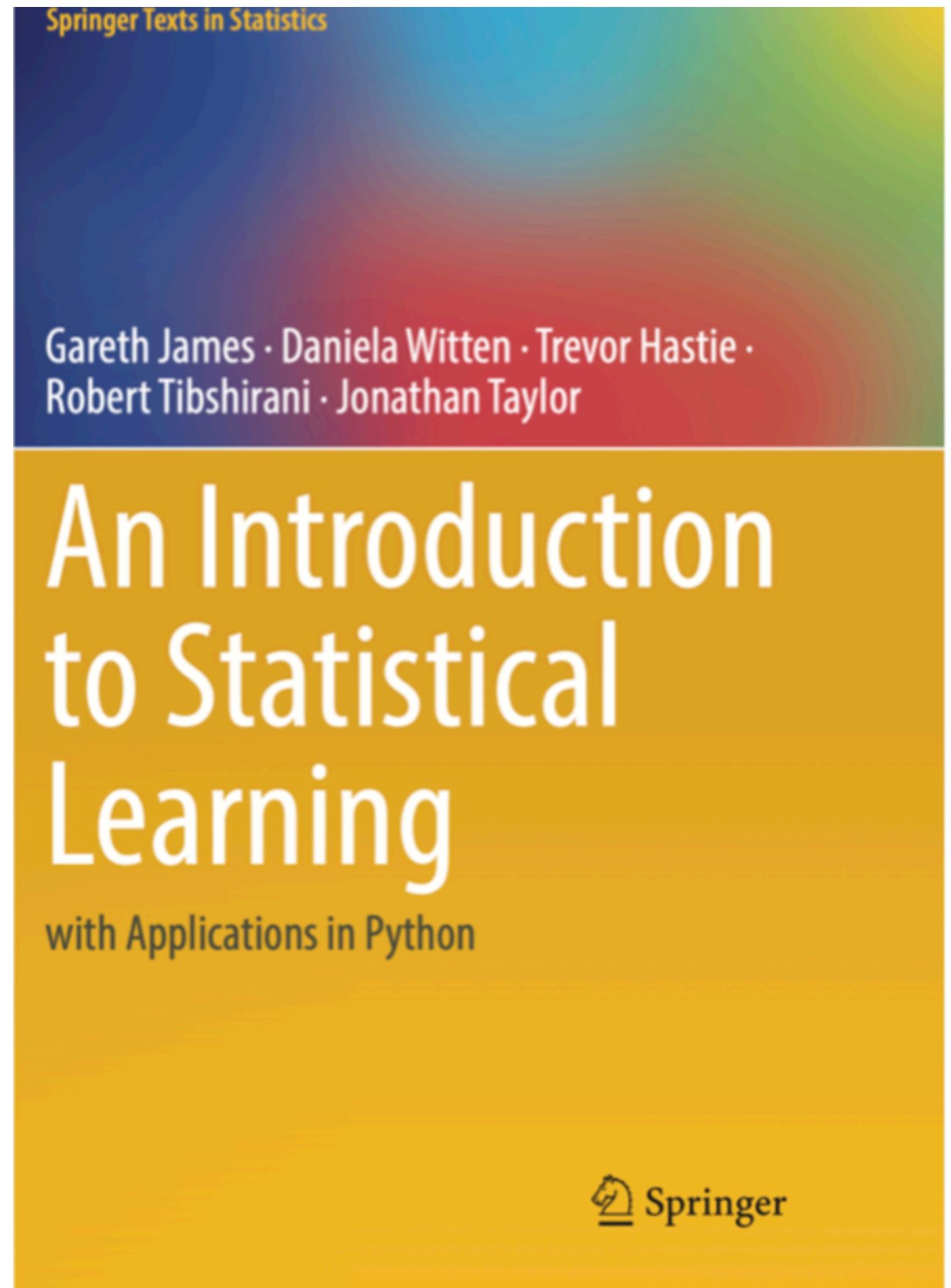
$$\leq \frac{1}{n} \lambda_1$$

- ▣ See 20261005-PCA-scratch.ipynb 🖱️



12.4

Clustering methods



<https://www.statlearning.com/>

Clustering

- *Clustering* refers to a very broad set of techniques for finding *subgroups*, or *clusters*, in a data set.
- We seek a partition of the data into distinct groups so that the observations within each group are quite similar to each other,
- To make this concrete, we must define what it means for two or more observations to be *similar* or *different*.
- Indeed, this is often a domain-specific consideration that must be made based on knowledge of the data being studied.

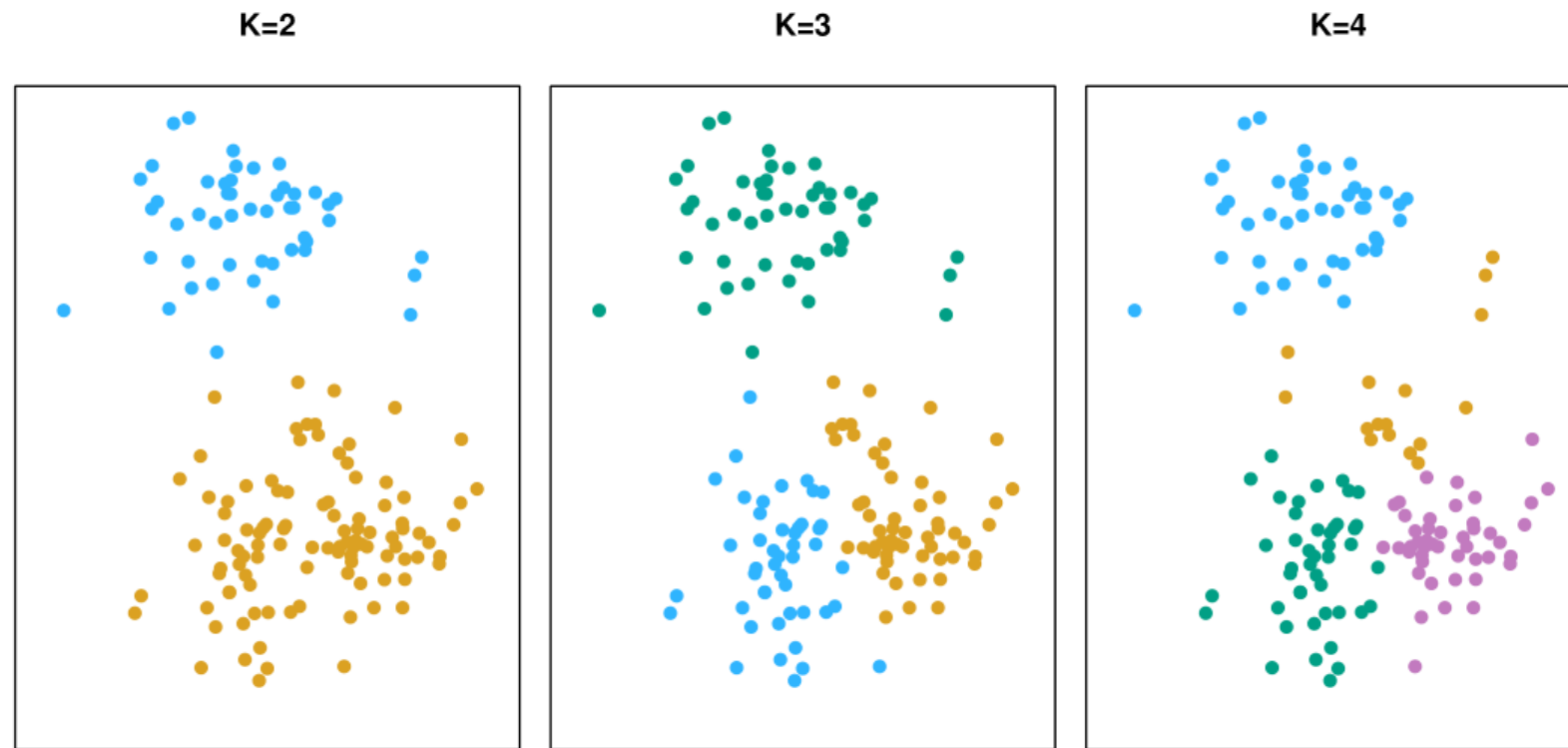
Clustering for Market Segmentation

- Suppose we have access to a large number of measurements (e.g. median household income, occupation, distance from nearest urban area, and so forth) for a large number of people.
- Our goal is to perform *market segmentation* by identifying subgroups of people who might be more receptive to a particular form of advertising, or more likely to purchase a particular product.
- The task of performing market segmentation amounts to clustering the people in the data set.

Two clustering methods

- In *K-means clustering*, we seek to partition the observations into a pre-specified number of clusters.
- In *hierarchical clustering*, we do not know in advance how many clusters we want; in fact, we end up with a tree-like visual representation of the observations, called a *dendrogram*, that allows us to view at once the clusterings obtained for each possible number of clusters, from 1 to n .

K -means clustering



A simulated data set with 150 observations in 2-dimensional space. Panels show the results of applying K -means clustering with different values of K , the number of clusters. The color of each observation indicates the cluster to which it was assigned using the K -means clustering algorithm. Note that there is no ordering of the clusters, so the cluster coloring is arbitrary. These cluster labels were not used in clustering; instead, they are the outputs of the clustering procedure.

Details of K -means clustering

Let $C_1, \dots, C_K$ denote sets containing the indices of the observations in each cluster. These sets satisfy two properties:

1. $C_1 \cup C_2 \cup \dots \cup C_K = \{1, \dots, n\}$. In other words, each observation belongs to at least one of the K clusters.
2. $C_k \cap C_{k'} = \emptyset$ for all $k \neq k'$. In other words, the clusters are non-overlapping: no observation belongs to more than one cluster.

For instance, if the i th observation is in the k th cluster, then $i \in C_k$.

Details of K -means clustering: continued

- The idea behind K -means clustering is that a *good* clustering is one for which the *within-cluster variation* is as small as possible.
- The within-cluster variation for cluster C_k is a measure $WCV(C_k)$ of the amount by which the observations within a cluster differ from each other.
- Hence we want to solve the problem

$$\text{minimize}_{C_1, \dots, C_K} \left\{ \sum_{k=1}^K WCV(C_k) \right\}. \quad (2)$$

- In words, this formula says that we want to partition the observations into K clusters such that the total within-cluster variation, summed over all K clusters, is as small as possible.

How to define within-cluster variation?

- Typically we use Euclidean distance

$$\text{WCV}(C_k) = \frac{1}{|C_k|} \sum_{i,i' \in C_k} \sum_{j=1}^p (x_{ij} - x_{i'j})^2, \quad (3)$$

where $|C_k|$ denotes the number of observations in the k th cluster.

- Combining (2) and (3) gives the optimization problem that defines K -means clustering,

$$\underset{C_1, \dots, C_K}{\text{minimize}} \left\{ \sum_{k=1}^K \frac{1}{|C_k|} \sum_{i,i' \in C_k} \sum_{j=1}^p (x_{ij} - x_{i'j})^2 \right\}. \quad (4)$$

K -Means Clustering Algorithm

1. Randomly assign a number, from 1 to K , to each of the observations. These serve as initial cluster assignments for the observations.
2. Iterate until the cluster assignments stop changing:
 - 2.1 For each of the K clusters, compute the cluster *centroid*.
The k th cluster centroid is the vector of the p feature means for the observations in the k th cluster.
 - 2.2 Assign each observation to the cluster whose centroid is closest (where *closest* is defined using Euclidean distance).

Properties of the Algorithm

- This algorithm is guaranteed to decrease the value of the objective (4) at each step. *Why?* Note that

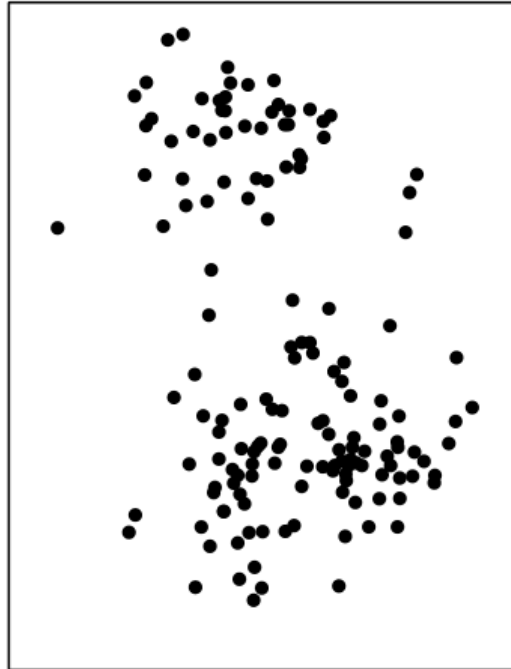
$$\frac{1}{|C_k|} \sum_{i,i' \in C_k} \sum_{j=1}^p (x_{ij} - x_{i'j})^2 = 2 \sum_{i \in C_k} \sum_{j=1}^p (x_{ij} - \bar{x}_{kj})^2,$$

where $\bar{x}_{kj} = \frac{1}{|C_k|} \sum_{i \in C_k} x_{ij}$ is the mean for feature j in cluster C_k .

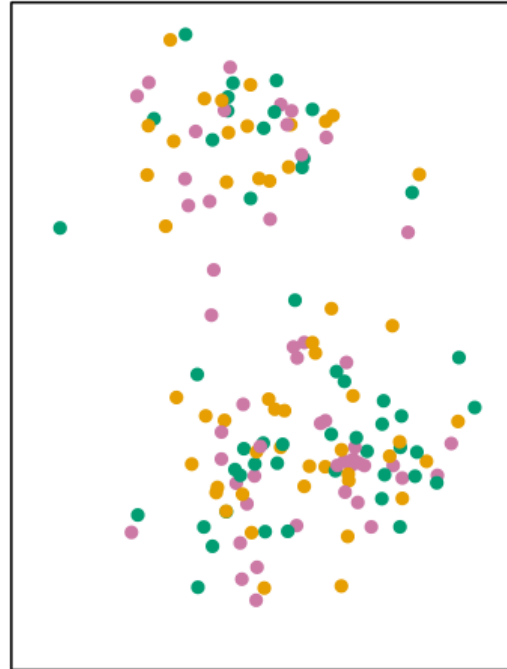
- however it is not guaranteed to give the global minimum.
Why not?

Example

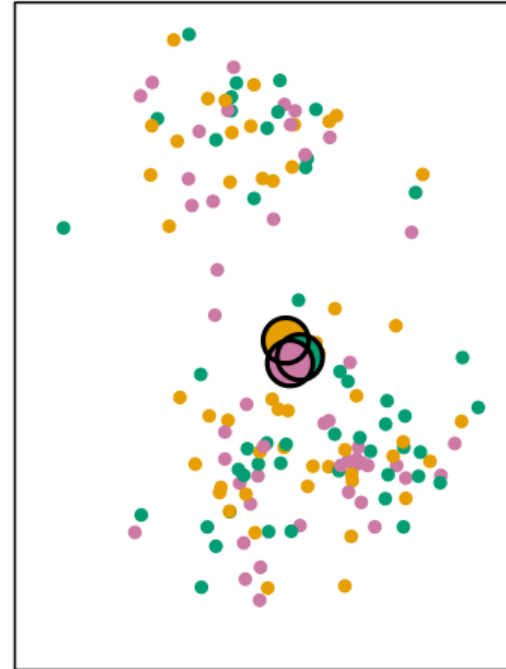
Data



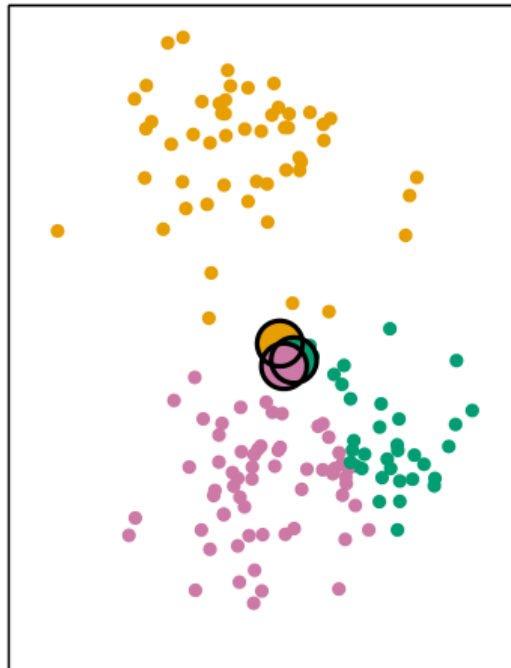
Step 1



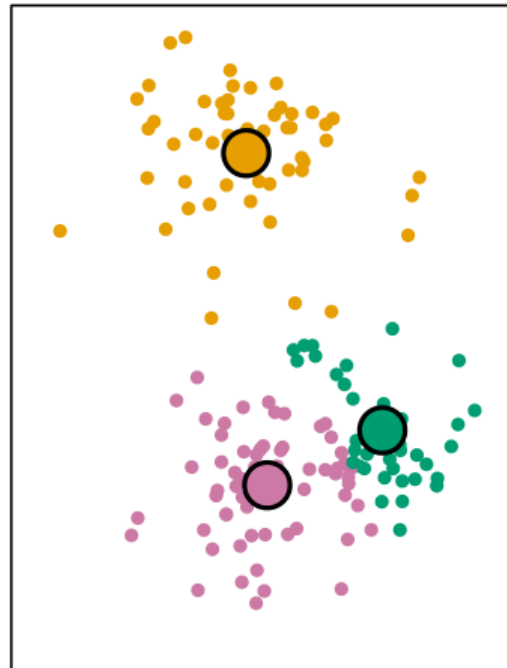
Iteration 1, Step 2a



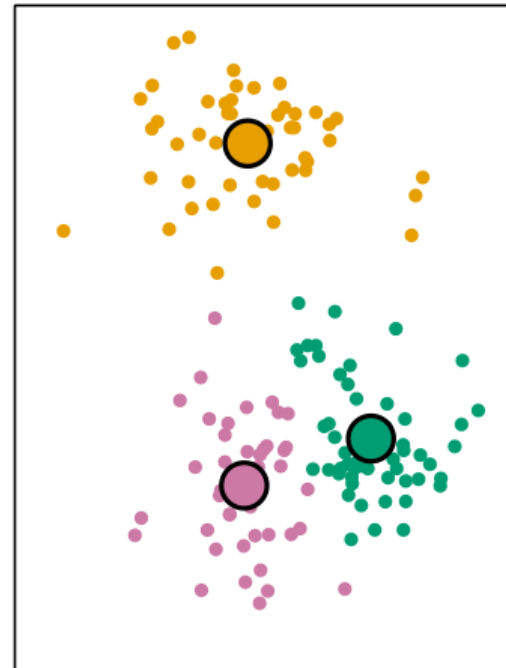
Iteration 1, Step 2b



Iteration 2, Step 2a



Final Results



Details of Previous Figure

The progress of the K-means algorithm with $K=3$.

- *Top left:* The observations are shown.
- *Top center:* In Step 1 of the algorithm, each observation is randomly assigned to a cluster.
- *Top right:* In Step 2(a), the cluster centroids are computed. These are shown as large colored disks. Initially the centroids are almost completely overlapping because the initial cluster assignments were chosen at random.
- *Bottom left:* In Step 2(b), each observation is assigned to the nearest centroid.
- *Bottom center:* Step 2(a) is once again performed, leading to new cluster centroids.
- *Bottom right:* The results obtained after 10 iterations.

Example: different starting values



Details of Previous Figure

K -means clustering performed six times on the data from previous figure with $K = 3$, each time with a different random assignment of the observations in Step 1 of the K -means algorithm.

Above each plot is the value of the objective (4).

Three different local optima were obtained, one of which resulted in a smaller value of the objective and provides better separation between the clusters.

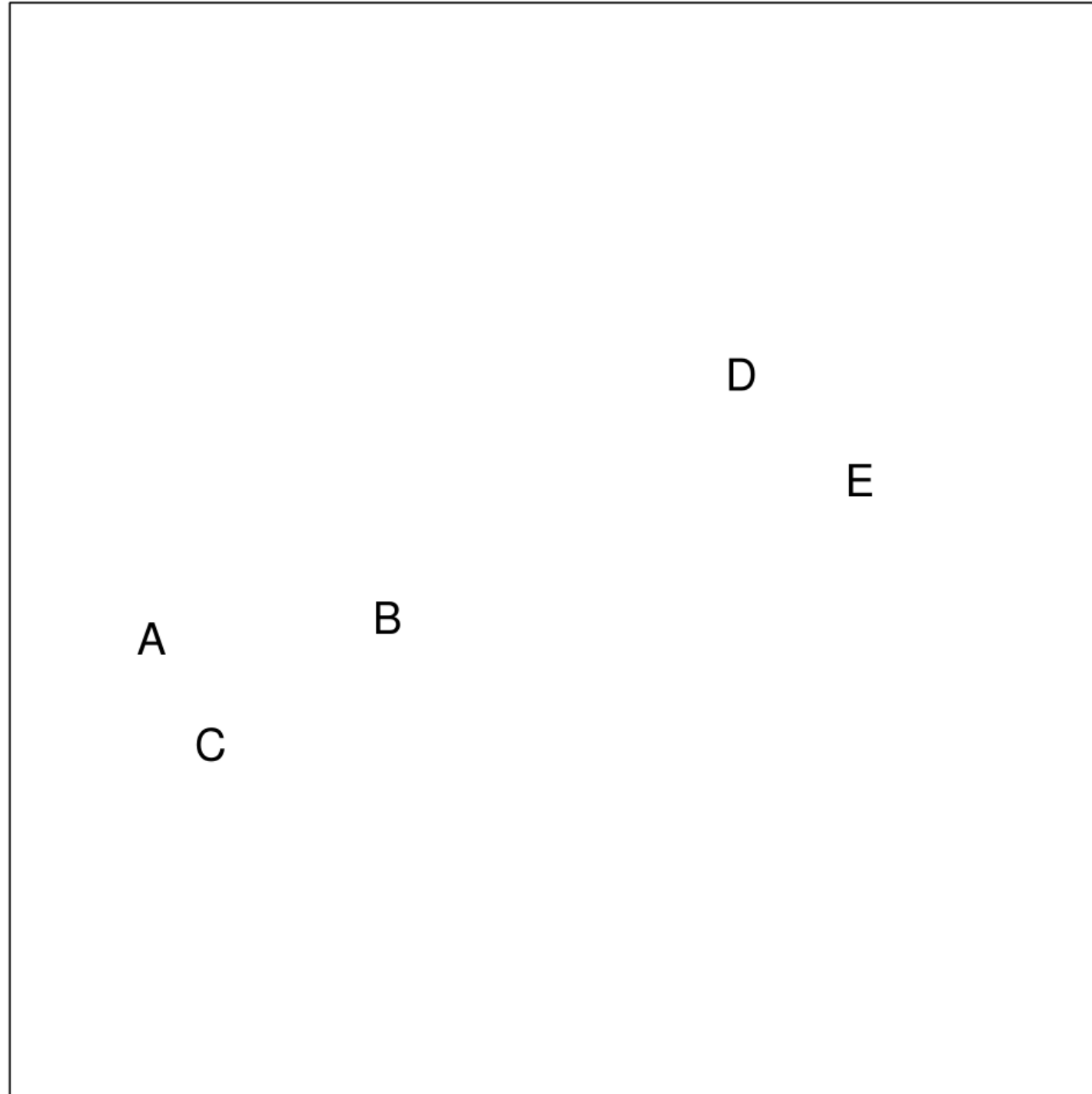
Those labeled in red all achieved the same best solution, with an objective value of 235.8

Hierarchical Clustering

- K -means clustering requires us to pre-specify the number of clusters K . This can be a disadvantage (later we discuss strategies for choosing K)
- *Hierarchical clustering* is an alternative approach which does not require that we commit to a particular choice of K .
- In this section, we describe *bottom-up* or *agglomerative* clustering. This is the most common type of hierarchical clustering, and refers to the fact that a dendrogram is built starting from the leaves and combining clusters up to the trunk.

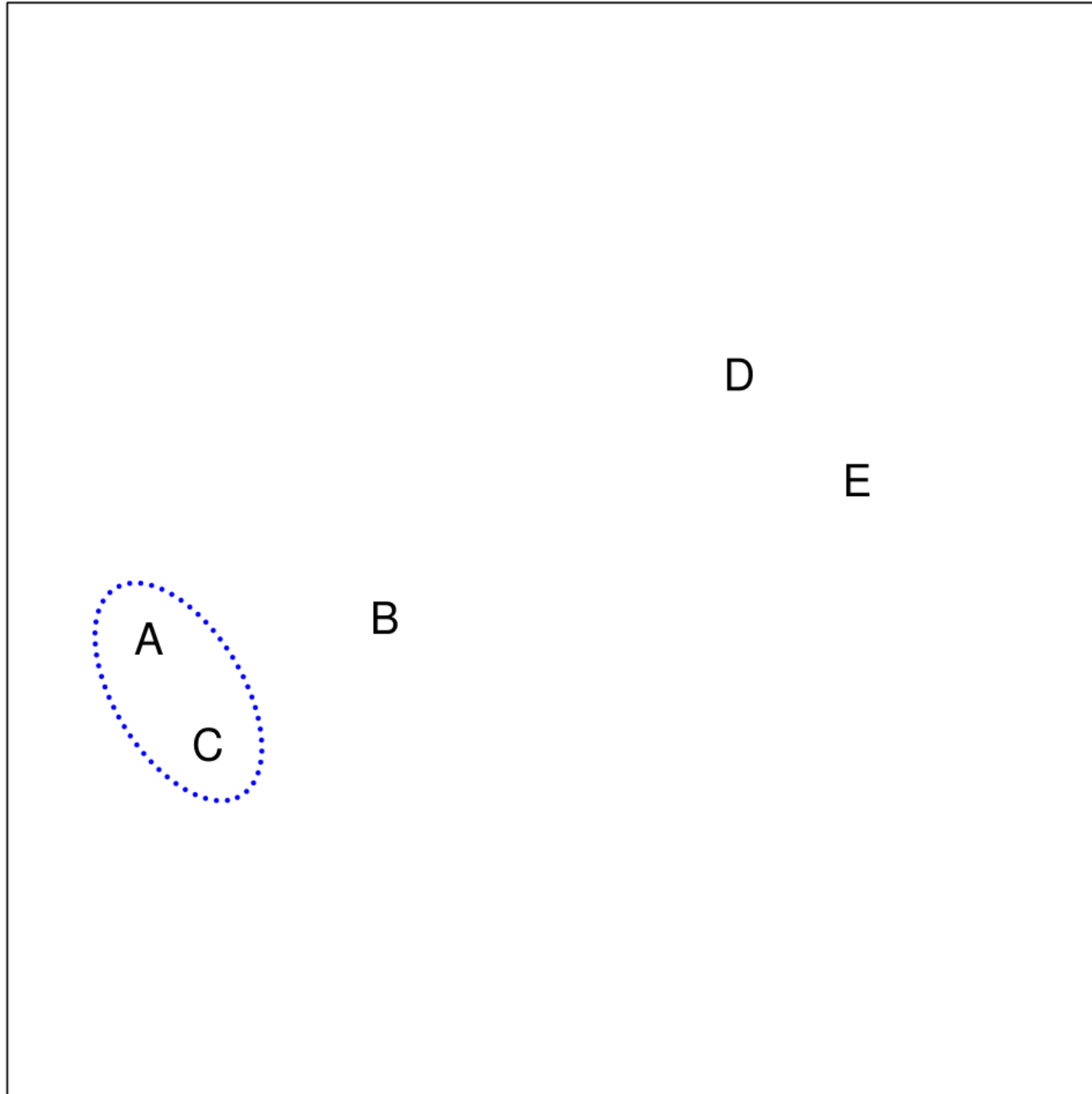
Hierarchical Clustering: the idea

Builds a hierarchy in a “bottom-up” fashion...



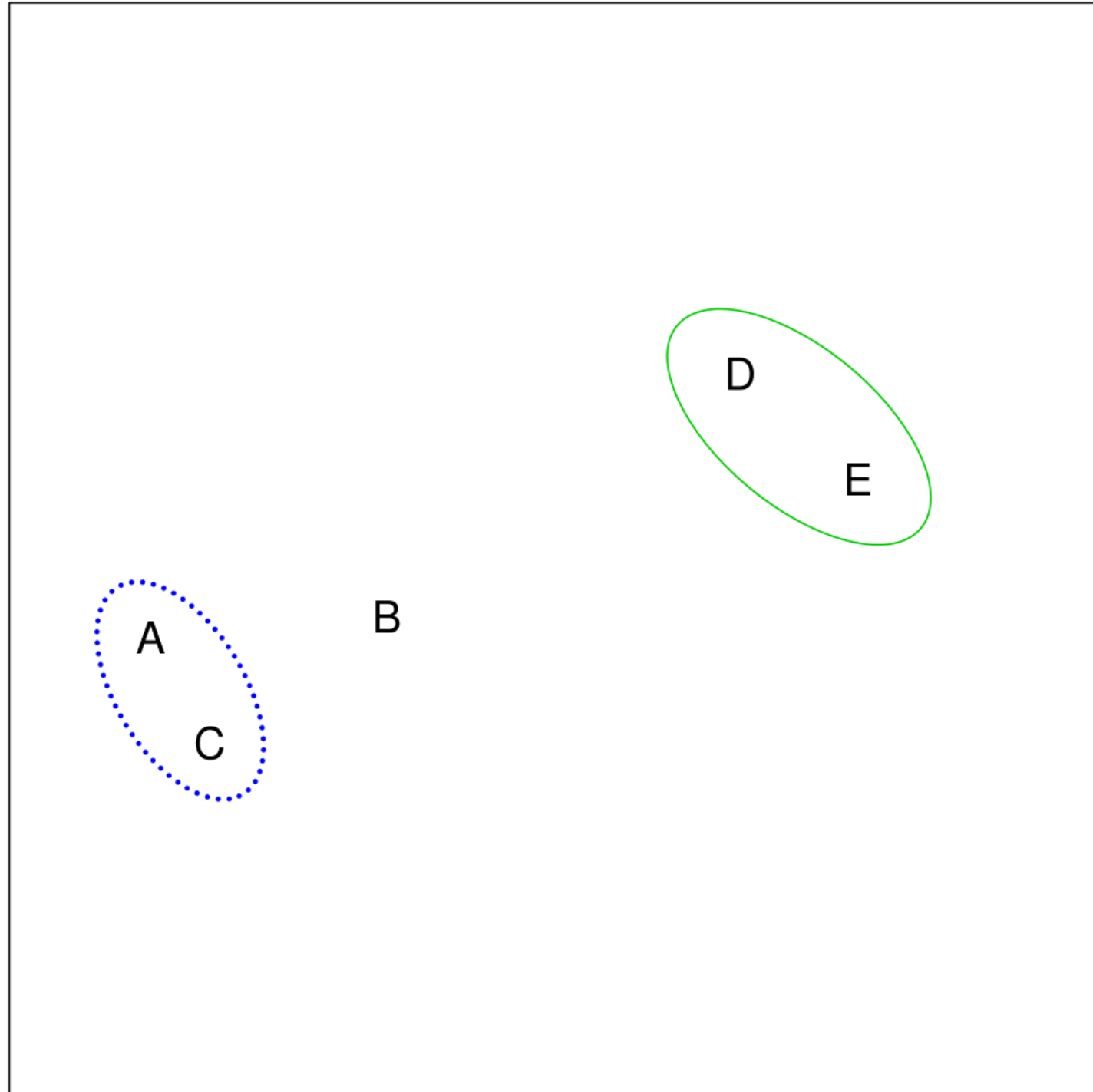
Hierarchical Clustering: the idea

Builds a hierarchy in a “bottom-up” fashion...



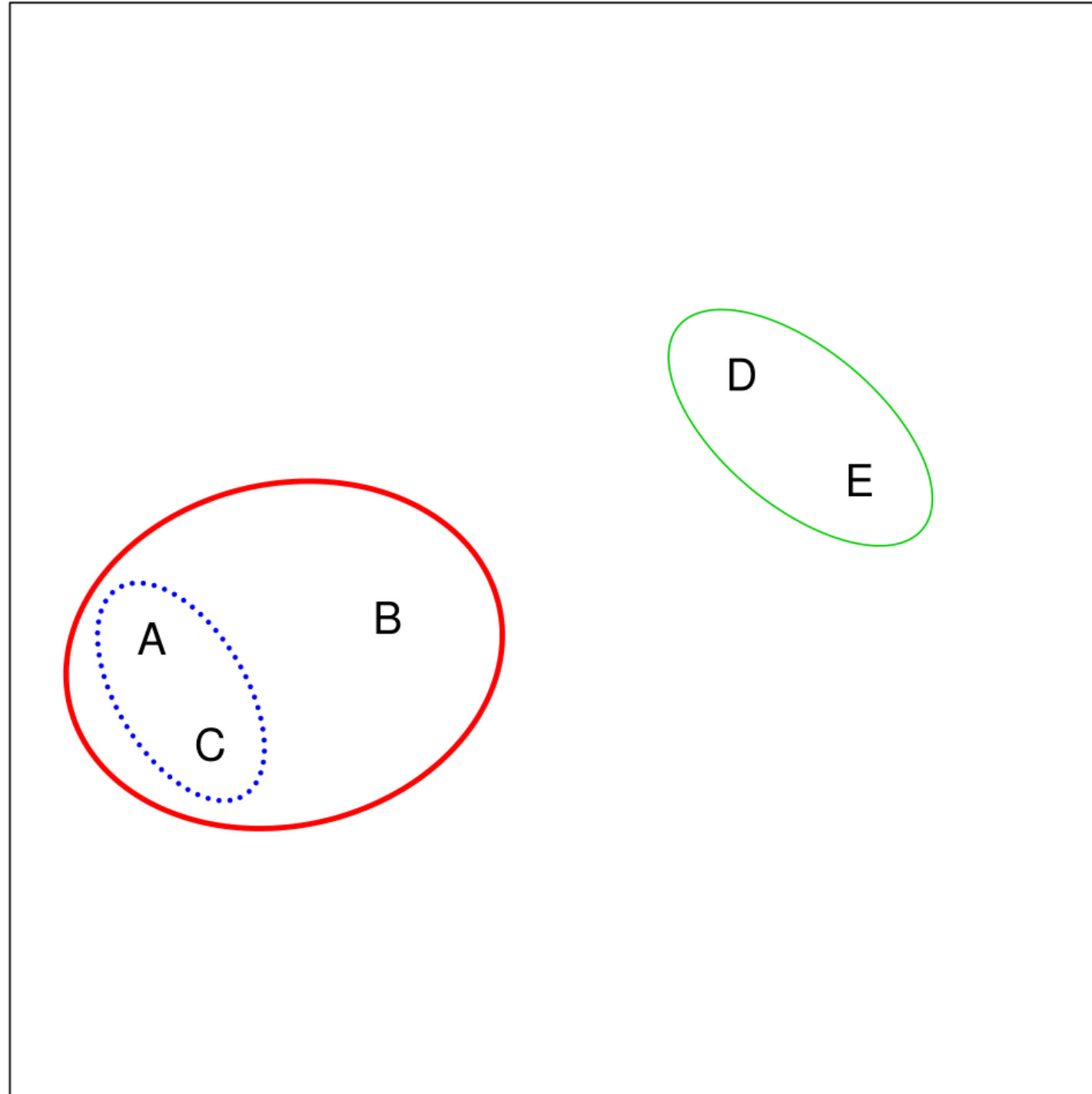
Hierarchical Clustering: the idea

Builds a hierarchy in a “bottom-up” fashion...



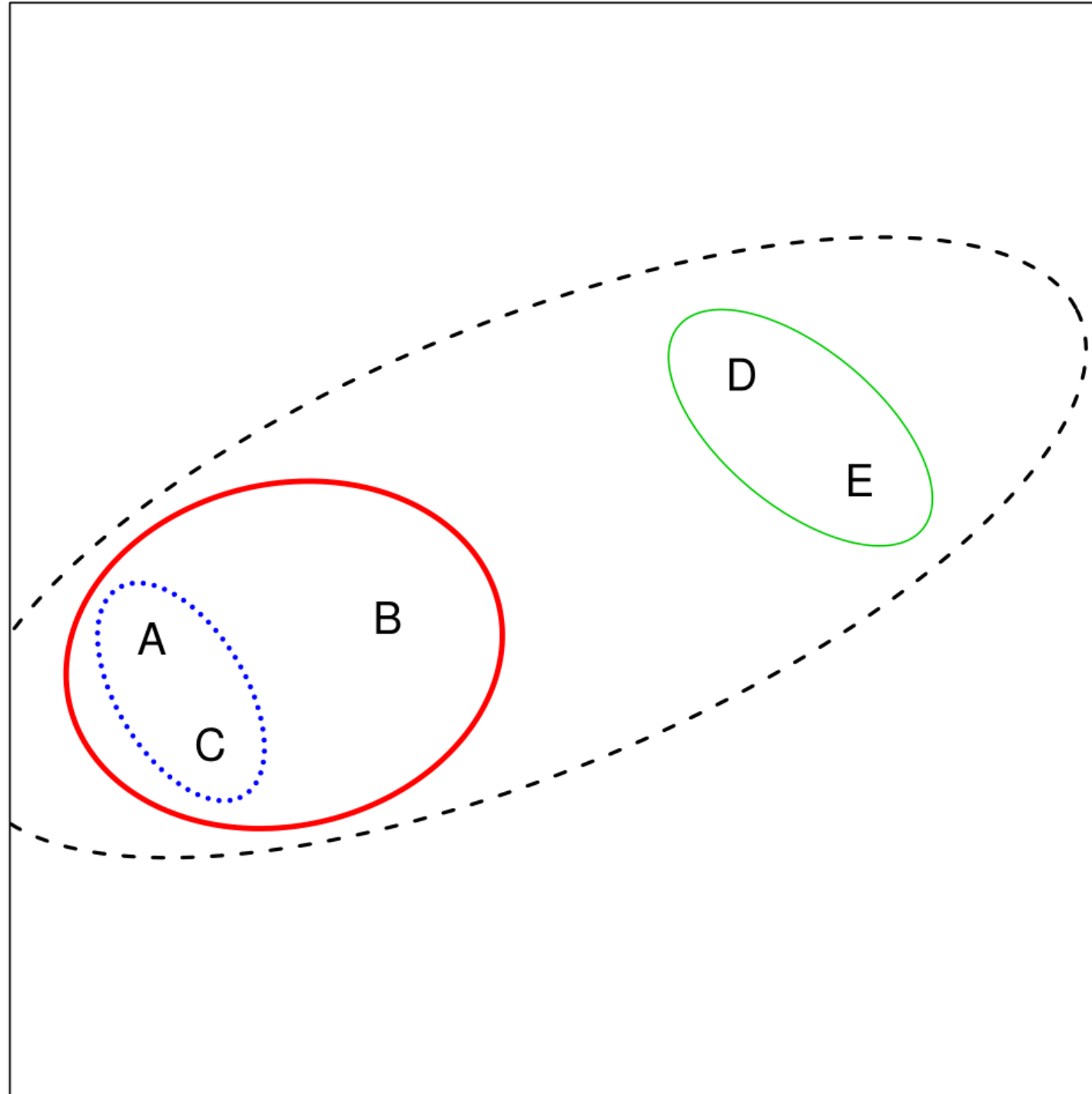
Hierarchical Clustering: the idea

Builds a hierarchy in a “bottom-up” fashion...



Hierarchical Clustering: the idea

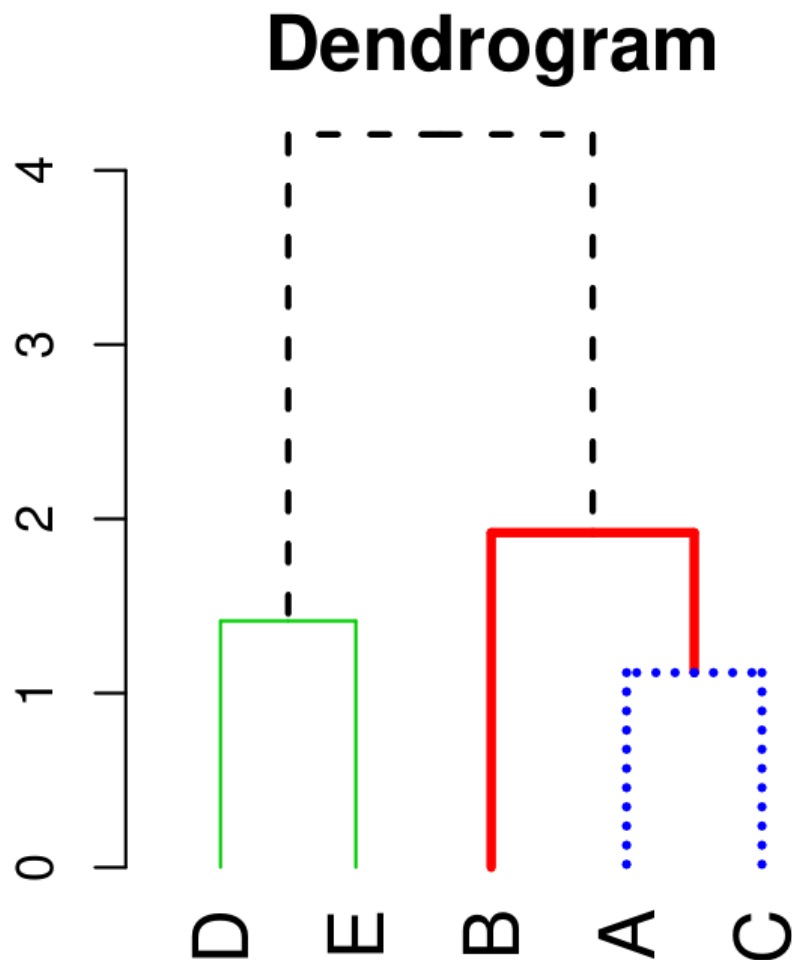
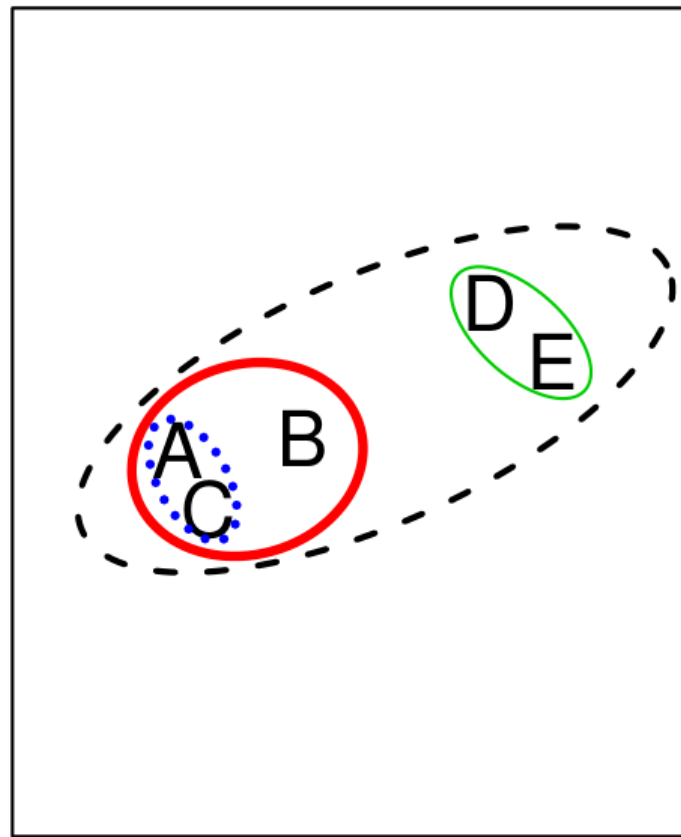
Builds a hierarchy in a “bottom-up” fashion...



Hierarchical Clustering Algorithm

The approach in words:

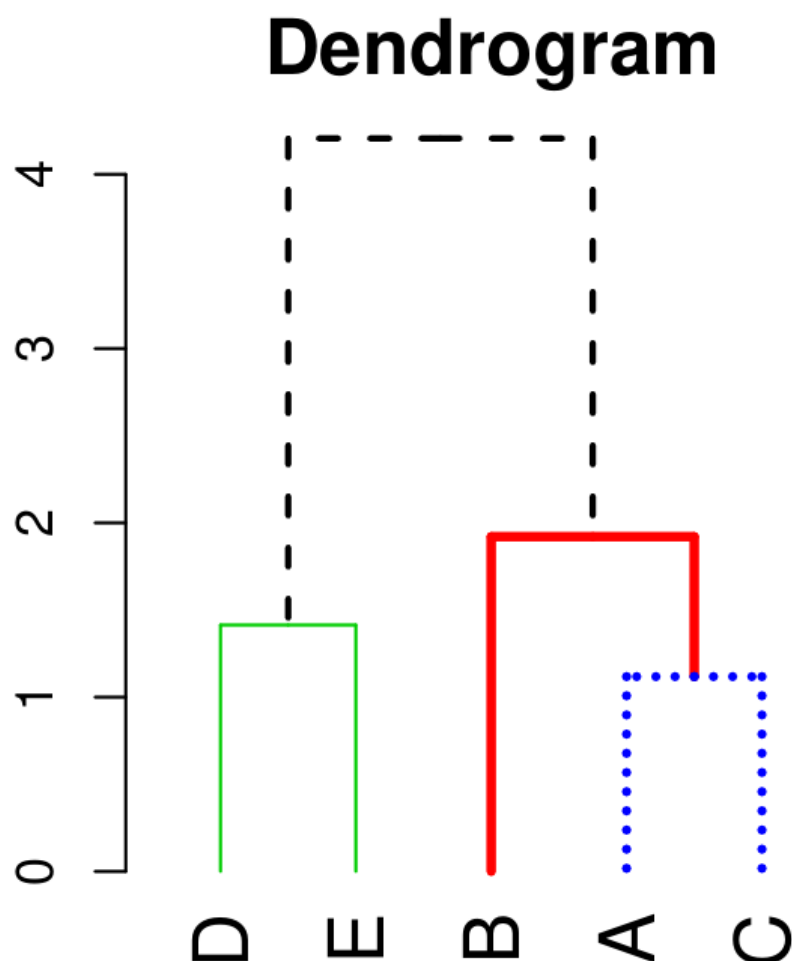
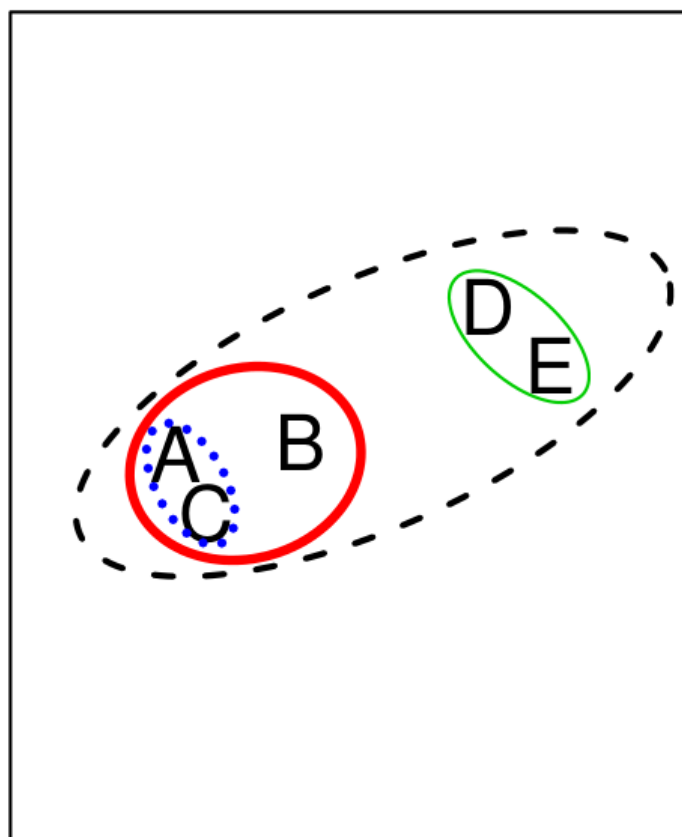
- Start with each point in its own cluster.
- Identify the closest two clusters and merge them.
- Repeat.
- Ends when all points are in a single cluster.



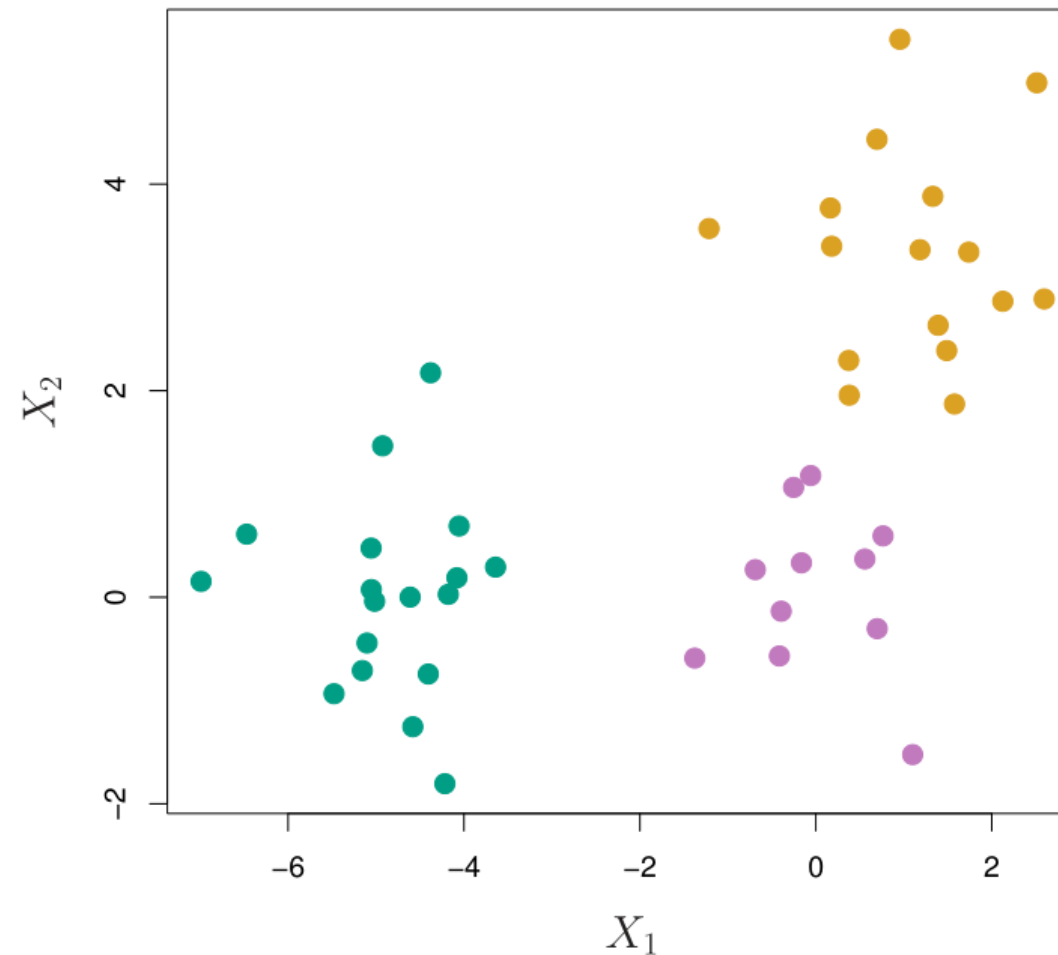
Hierarchical Clustering Algorithm

The approach in words:

- Start with each point in its own cluster.
- Identify the **closest** two clusters and merge them.
- Repeat.
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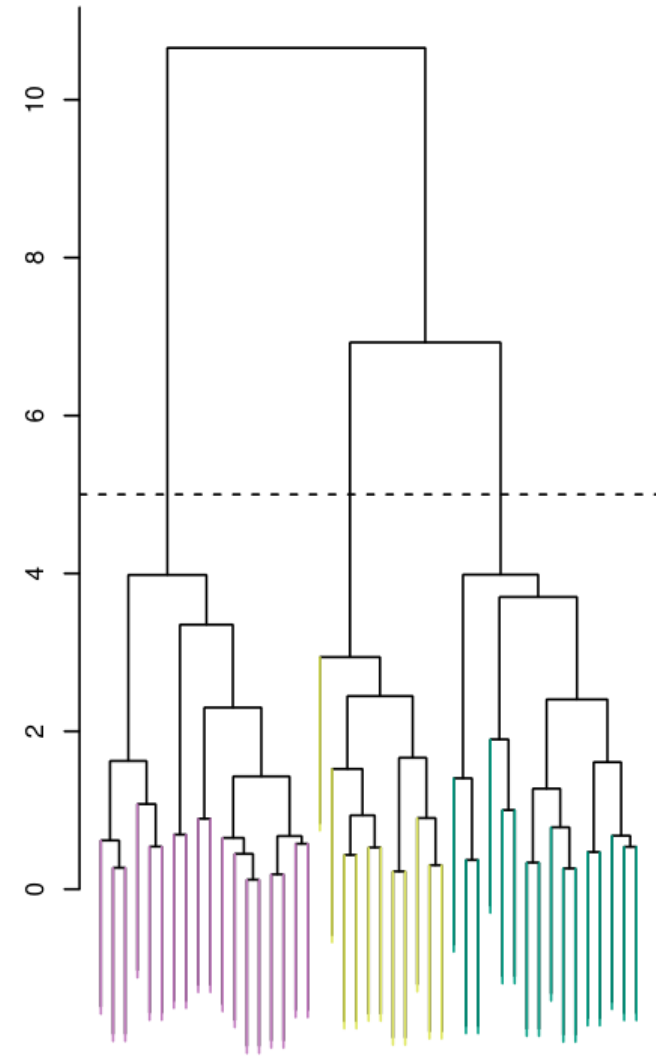
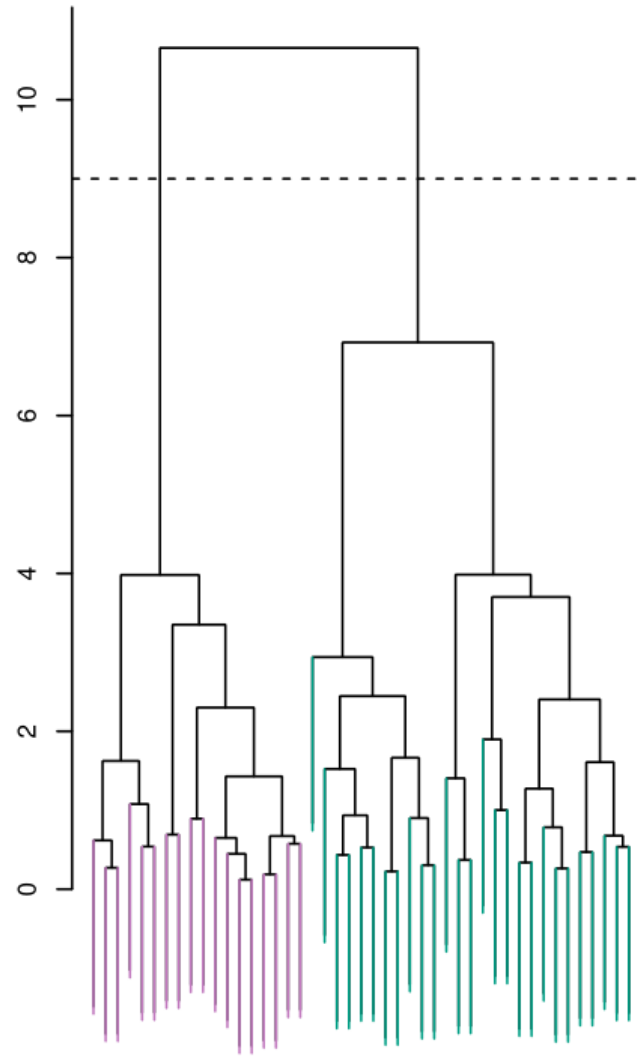
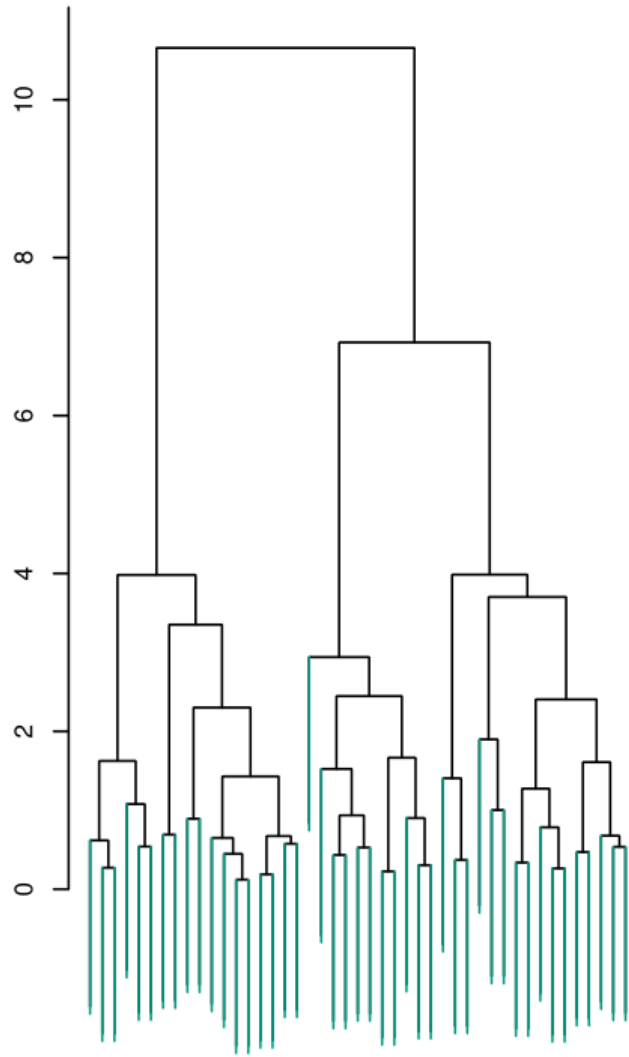


An Example



45 observations generated in 2-dimensional space. In reality there are three distinct classes, shown in separate colors. However, we will treat these class labels as unknown and will seek to cluster the observations in order to discover the classes from the data.

Application of hierarchical clustering



Details of previous figure

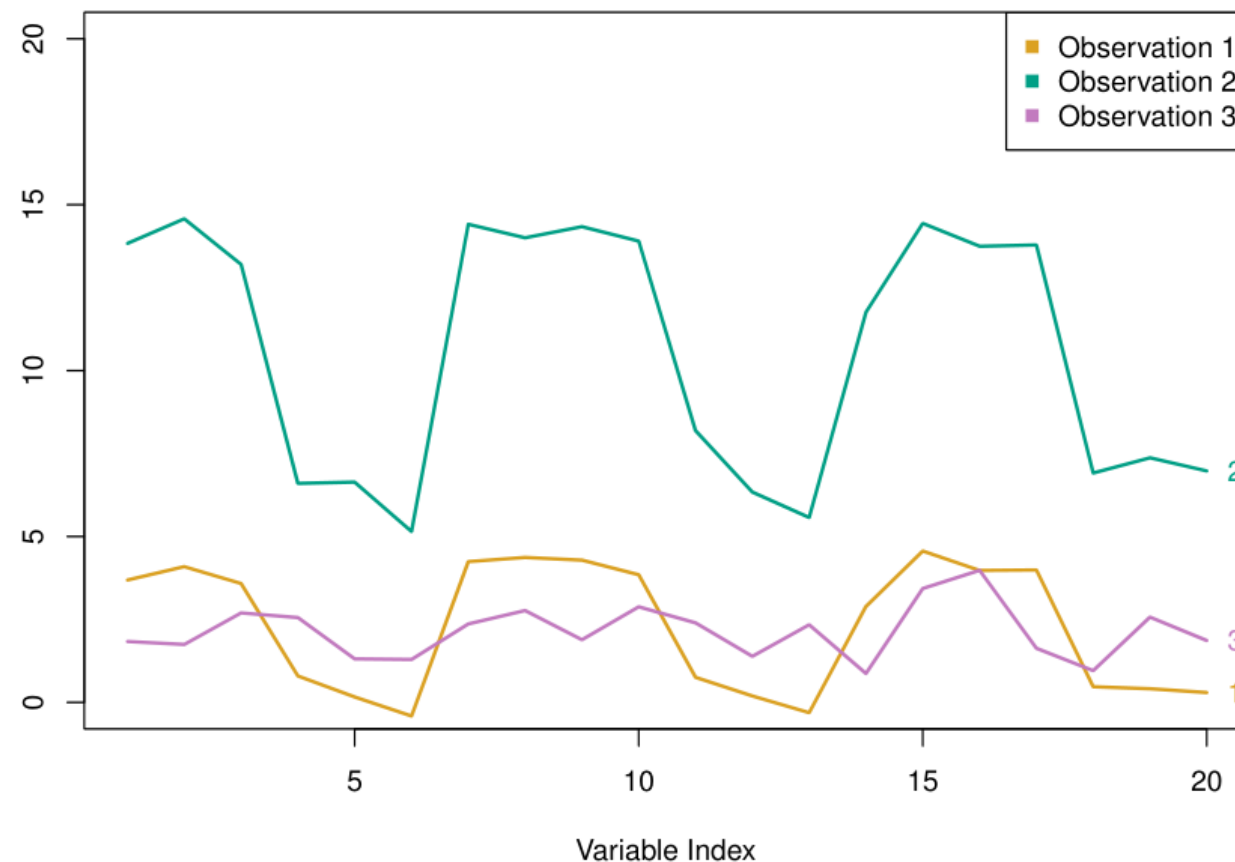
- *Left:* Dendrogram obtained from hierarchically clustering the data from previous slide, with complete linkage and Euclidean distance.
- *Center:* The dendrogram from the left-hand panel, cut at a height of 9 (indicated by the dashed line). This cut results in two distinct clusters, shown in different colors.
- *Right:* The dendrogram from the left-hand panel, now cut at a height of 5. This cut results in three distinct clusters, shown in different colors. Note that the colors were not used in clustering, but are simply used for display purposes in this figure

Types of Linkage

<i>Linkage</i>	<i>Description</i>
Complete	Maximal inter-cluster dissimilarity. Compute all pairwise dissimilarities between the observations in cluster A and the observations in cluster B, and record the <i>largest</i> of these dissimilarities.
Single	Minimal inter-cluster dissimilarity. Compute all pairwise dissimilarities between the observations in cluster A and the observations in cluster B, and record the <i>smallest</i> of these dissimilarities.
Average	Mean inter-cluster dissimilarity. Compute all pairwise dissimilarities between the observations in cluster A and the observations in cluster B, and record the <i>average</i> of these dissimilarities.
Centroid	Dissimilarity between the centroid for cluster A (a mean vector of length p) and the centroid for cluster B. Centroid linkage can result in undesirable <i>inversions</i> .

Choice of Dissimilarity Measure

- So far have used Euclidean distance.
- An alternative is *correlation-based distance* which considers two observations to be similar if their features are highly correlated.
- This is an unusual use of correlation, which is normally computed between variables; here it is computed between the observation profiles for each pair of observations.



Practical issues

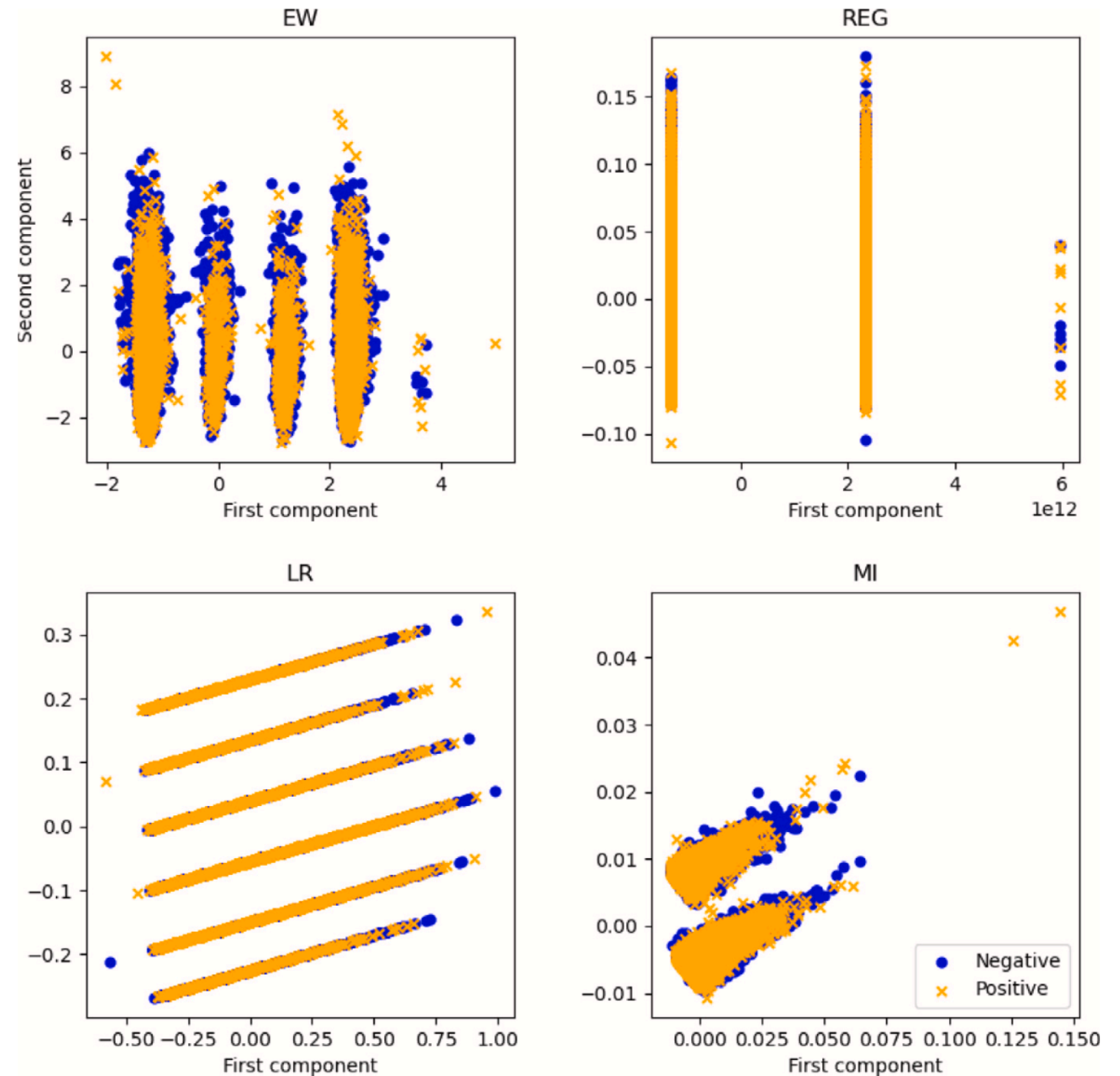
- *Scaling of the variables matters!*. Should the observations or features first be standardized in some way? For instance, maybe the variables should be centered to have mean zero and scaled to have standard deviation one.
- In the case of hierarchical clustering,
 - What dissimilarity measure should be used?
 - What type of linkage should be used?
- How many clusters to choose? (in both K -means or hierarchical clustering). Difficult problem. No agreed-upon method. See Elements of Statistical Learning, chapter 13 for more details.
- Which features should we use to drive the clustering?



Bridging accuracy and interpretability: A rescaled cluster-then-predict approach for enhanced credit scoring

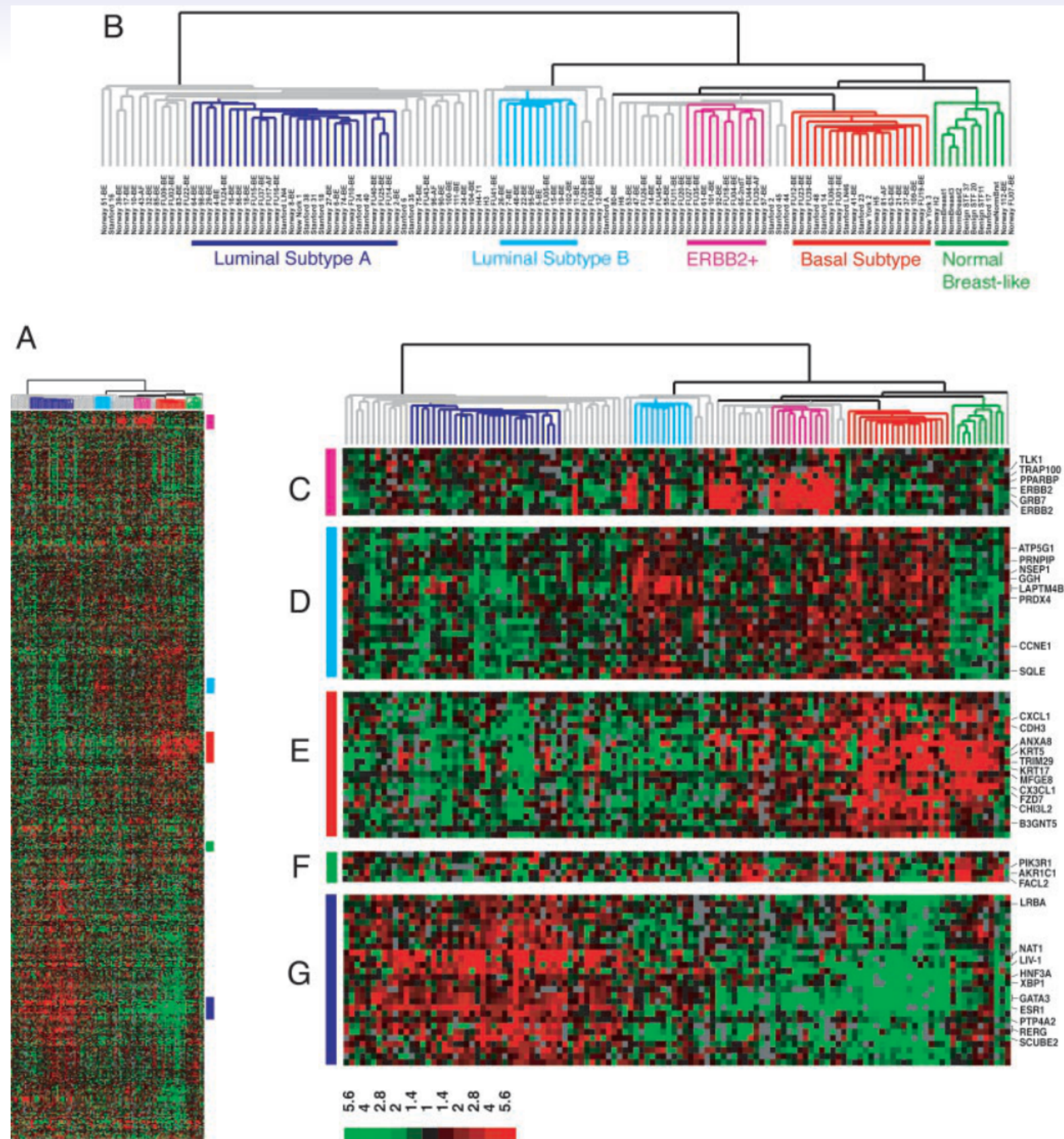
Huei-Wen Teng^{*}, Ming-Hsuan Kang, I-Han Lee, Le-Chi Bai

National Yang Ming Chiao Tung University, No 1001 Daxue Road East District, Hsinchu City, 300093, Taiwan, ROC

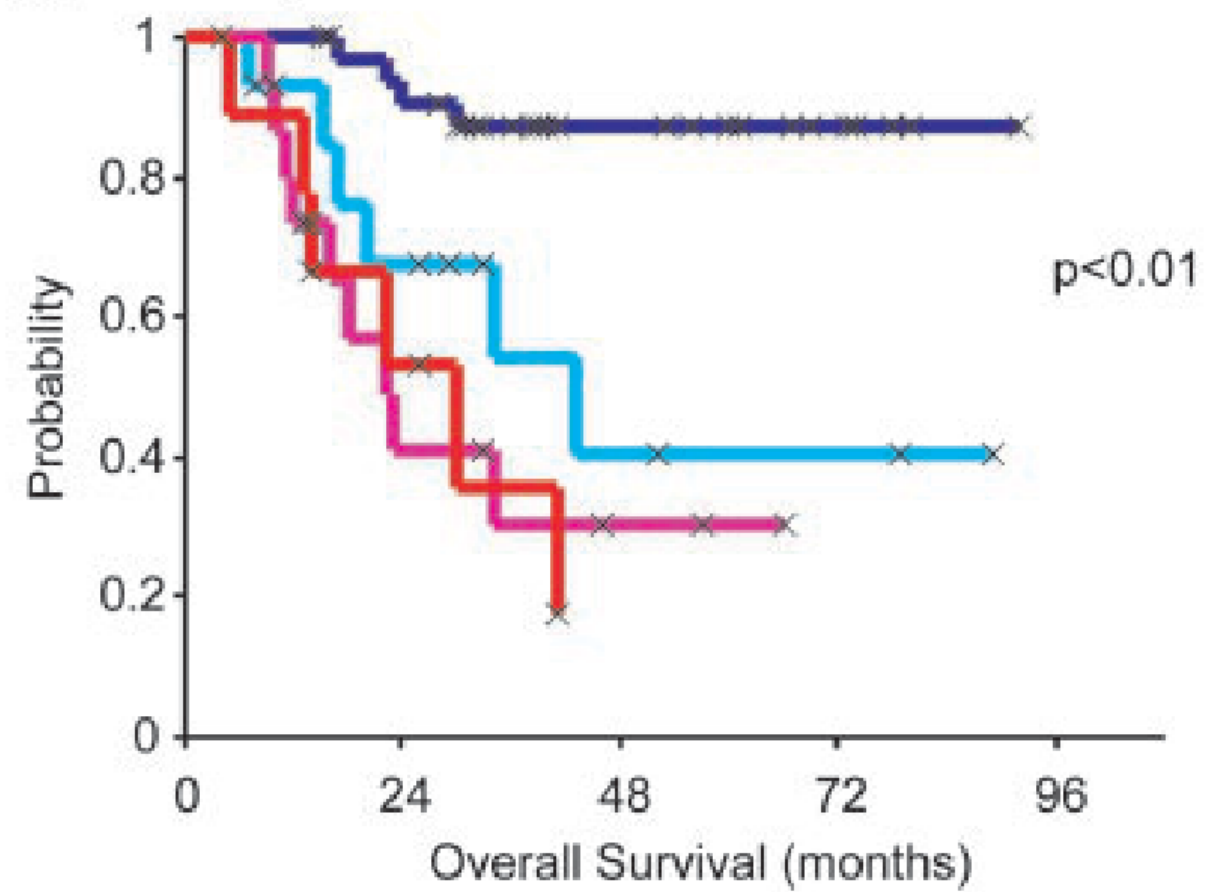


Example: breast cancer microarray study

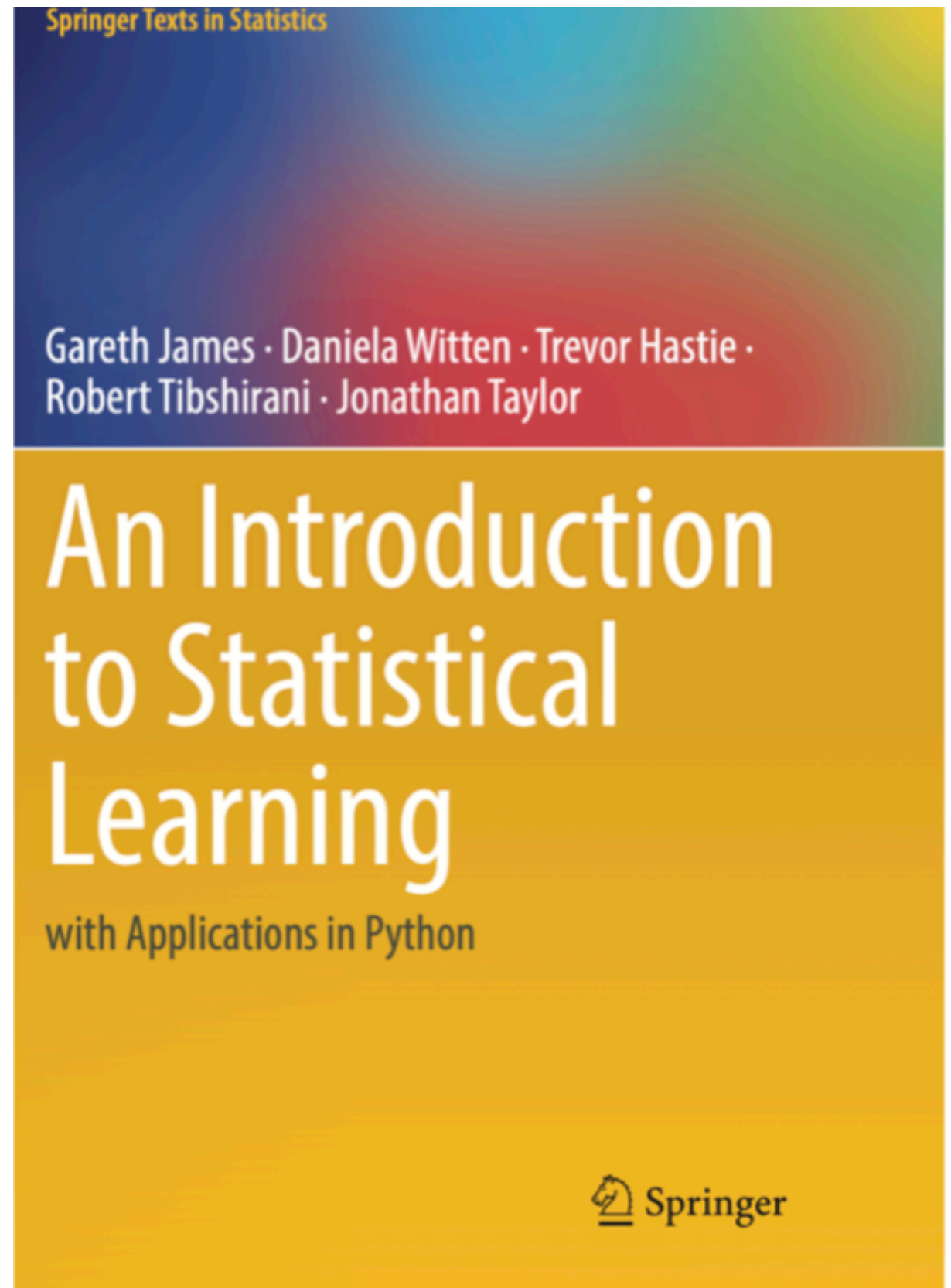
- “Repeated observation of breast tumor subtypes in independent gene expression data sets;” Sorlie et al, PNAS 2003
- Gene expression measurements for about ~ 8000 genes, for each of 88 breast cancer patients.
- Average linkage, correlation metric
- Clustered samples using 500 *intrinsic genes*: each woman was measured before and after chemotherapy. Intrinsic genes have smallest within/between variation.



B Norway/Stanford data set



12.5 Lab: Unsupervised learning



<https://www.statlearning.com/>

Conclusions

- *Unsupervised learning* is important for understanding the variation and grouping structure of a set of unlabeled data, and can be a useful pre-processor for supervised learning
- It is intrinsically more difficult than *supervised learning* because there is no gold standard (like an outcome variable) and no single objective (like test set accuracy).
- It is an active field of research, with many recently developed tools such as *self-organizing maps*, *independent components analysis* and *spectral clustering*.
See *The Elements of Statistical Learning*, chapter 14.